# The Briefing: something is coming

**LEARNING TARGET · R1** I can describe what we know and don't know about 2026-XJ, and name the physics tools we'll need.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Observation** | Something we record or measure, like a position at a time. |
| **Model** | A useful "if": *if* the motion keeps going the same way, we can continue the pattern. |
| **Prediction** | An estimate of what comes next, made by continuing the pattern. |
| **Testable question** | One that measurements can answer. "Where will it be tomorrow?" — yes. "Who should pay for a mission?" — no; that also needs human decisions. |

## SECTION B UNIT 1 ANCHOR + LANGUAGE OF MOTION

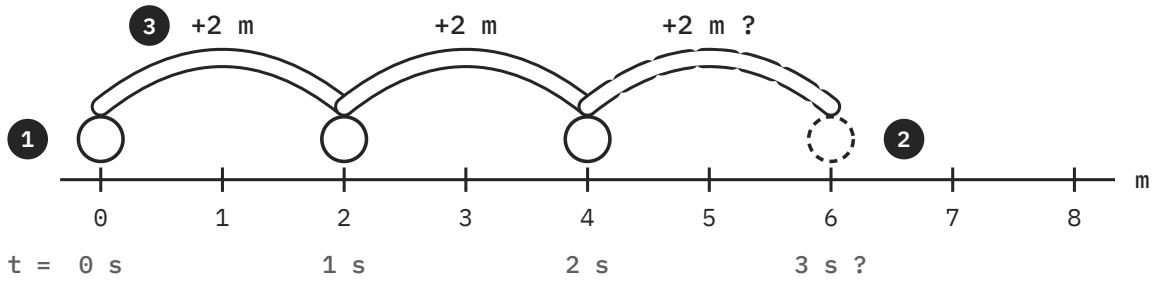
**How fast, and where will it arrive?**

- **x** position: where it is
- **v** velocity: how fast + which way
- **a** acceleration: how v changes

how fast · where · when · how big · made of · stop it?

## SECTION C COLOR IT · WATCH A PATTERN FORM

*One dot marks the cart's position each second.*




- ☐ **1 · BLUE** Observations: measured positions
- ☐ **2 · BLUE** Prediction: dashed outline
- ☐ **3 · PURPLE** The pattern: +2 m each second

## SECTION D WORKED EXAMPLE

A cart is at 2 m, 3 m, then 4 m, one second apart. Predict its position one second later.

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| x = 2, 3, 4 m<br>1 s between readings | next = last + change each second | change = 3 − 2 = +1 m<br>next = 4 m + 1 m | **5 m**<br>if the motion continues the same way |

## SECTION E THE 2026-XJ THREAD

DAY 1 OF 22 · NEW TOOL: THE UNIT QUESTION

The desk has a sighting, not a forecast. Everything we learn in Unit 1 serves one question: how fast is 2026-XJ moving, and where will it arrive?

*Unit question: How fast is it moving, and where will it arrive?*

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Start with what we can observe

Observations, models, and a prediction we can test

Suppose we learn that an object in space is approaching Earth. That gives us something to investigate, but it does not tell us where the object will arrive. Approaching Earth and hitting Earth are different claims. Our job is to use observations to build a prediction we can explain. In this unit, we will do that with our fictional asteroid, 2026-XJ.

Find observation in **Section A**. An observation is something we record or measure. A position at a particular time is an observation. A prediction is an estimate of what comes next. To connect the two, we use a model: a useful description with an assumption. We might say, “If the motion keeps going the same way, this pattern will continue.” The word if matters because the next observation can test that assumption.

Look at the position dots in **Section C**. A cart is at **0 m, 2 m, and 4 m**, with one second between measurements. Each second its position increases by 2 m. If the pattern continues, we predict **6 m at 3 s**. The dashed outline marks a prediction, not another measurement. Blue identifies position. Purple identifies the motion pattern connecting those positions.

**Section D** uses a different cart. Its measured positions are **2 m, 3 m, and 4 m**, again one second apart. First find what changes between readings: **3 m − 2 m = +1 m**, and **4 m − 3 m = +1 m**. The two changes agree. That gives us a reason to extend the pattern by one more second.

Our rule is **next position = last position + change during the next second**. Applying it gives **4 m + 1 m = 5 m**. I want the condition to stay with the answer: the prediction is 5 m if the motion continues the same way. If a new reading gives a different position, we should check the measurement and whether the cart changed its motion.

Three symbols will help us describe motion throughout the unit. Position, $x$, tells us where an object is relative to a chosen zero. Velocity, $v$, tells us how fast position changes and which way the object moves. Acceleration, $a$, tells us how velocity changes. These quantities answer different questions. We will learn how to connect them instead of treating every number as just “how fast.”

A testable question is one that measurements can help answer. “Where will it be tomorrow?” is testable once we have enough information and a model. “Who should pay for a mission?” also involves human priorities and decisions. Physics can inform those decisions, but a calculation alone cannot make them.

For 2026-XJ, we will need positions, times, directions, and information about how reliable the measurements are. Later, forces will help us explain changes in its motion. The question tying that work together is the one on your notes page: how fast is it moving, and where will it arrive?

# Talking about motion

**LEARNING TARGET · K1** I can explain how position, distance, and displacement differ.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Position** | *Where are you?* Measured from a chosen zero, with a chosen + direction. |
| **Distance** | *How much travel?* Counts every part of the path. Never negative. |
| **Displacement** | *What changed from start to finish?* The sign gives the direction. |
| **Scalar / vector** | Distance has size only (scalar). Displacement has size *and* direction (vector). |

## SECTION B KEY EQUATIONS + UNITS

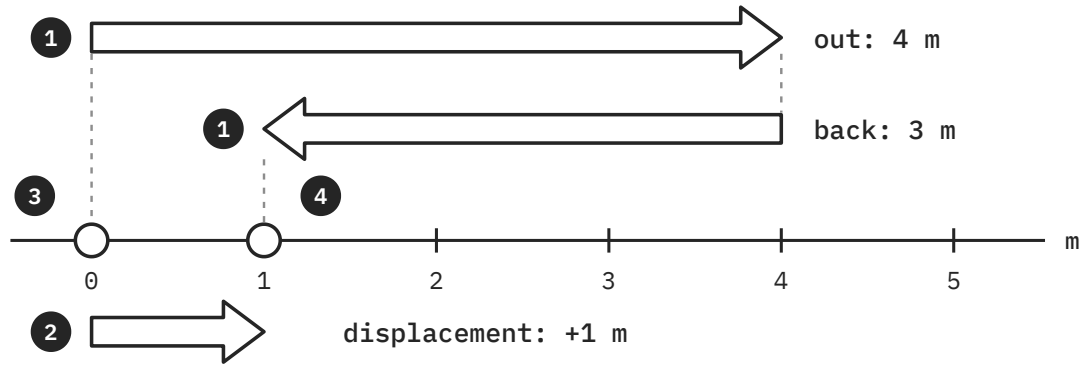
$$\Delta x = x_f - x_i$$

distance = leg 1 + leg 2 + …

- $\Delta$ "change in" (finish − start)
- $x$ position, meters (m)

## SECTION C COLOR IT · ONE TRIP, THREE QUESTIONS

*0 m → 4 m → 1 m. Right is +.*




- ☐ **1 · PENCIL** Distance: trace both legs
- ☐ **2 · BLUE** Displacement arrow
- ☐ **3 · BLUE** Start position
- ☐ **4 · BLUE** Finish position

## SECTION D WORKED EXAMPLE

A cart goes from 2 m to 6 m, then back to 4 m. Right is positive. Find the distance and the displacement.

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| $x_i = 2$ m<br>turn at 6 m<br>$x_f = 4$ m | distance = leg 1 + leg 2<br>$\Delta x = x_f - x_i$ | 4 m + 2 m<br>4 m − 2 m | distance **6 m**<br>displacement **+2 m** |

## SECTION E THE 2026-XJ THREAD

DAY 2 OF 22 · NEW TOOL: A REFERENCE FRAME

Before we can say where XJ is, we need a zero and a + direction. Analysts measure its position from Earth and track its displacement, not the winding path it travels.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Where you finish is not the whole trip

Position, distance, and displacement

If I say a cart is at two meters, you need to know where we started measuring. Position needs a chosen zero, called the origin. We also choose a positive direction. On today's notes page, right is positive. Once those choices are clear, a position tells us where the cart is along the line.

Position is a location. Distance counts how much travel occurred. Displacement describes the change from starting position to final position. Those are three different questions, so one trip can have three different descriptions. I want you to decide which question is being asked before calculating anything.

Trace both legs in **Section C**. It shows a trip from **0 m to 4 m, then back to 1 m**. The outward leg is four meters long. The return leg is three meters long. Total distance is **distance = 4 m + 3 m = 7 m**. Coming back does not erase the first leg. Distance counts every part of the route and is never negative.

Displacement uses only the starting and finishing positions. The symbol $\Delta$ means "change in," so $\Delta x$ means change in position. The rule is $\boldsymbol{\Delta x = x_f - x_i}$: finish minus start. For the diagram, $\boldsymbol{\Delta x}$ **= 1 m − 0 m = +1 m**. The cart finishes one meter right of where it began, even though it traveled seven meters.

Now find the starting and turning positions in **Section D**. The cart starts at **2 m**, travels to **6 m**, and returns to **4 m**. Its first leg is **6 m − 2 m = 4 m**. The return leg has length **6 m − 4 m = 2 m**. Add the lengths to find **distance = 4 m + 2 m = 6 m**. The direction of a leg does not make its length negative.

For displacement, use the endpoints instead: $\boldsymbol{\Delta x}$ **= 4 m − 2 m = +2 m**. The final position is 4 m from our chosen zero, while the displacement is 2 m right of the starting point. Those answers differ because the cart did not start at zero. This is a common place to lose track of what a number means.

Distance is a scalar: it has size and units. Displacement is a vector: it also has direction. Along one axis, the sign gives that direction. A negative displacement would mean a finish left of the start. In the diagram, the traveled route stays in pencil; the positions and displacement arrow are blue.

For our fictional asteroid, 2026-XJ, we need the same clear reference frame. Measuring from Earth gives our classroom model a zero, but we must still name the positive direction. Total distance describes the route traveled. Starting position and displacement tell us where the asteroid ends up. That is the information a position forecast needs.

# Velocity

**LEARNING TARGET · S1** I can use a position–time graph's slope to find velocity.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Velocity** | *How fast and which way.* The rate position changes. |
| **Speed** | *How fast only.* The size of the velocity, with no direction. |
| **Slope** | Steepness of a line: rise ÷ run. On a position–time graph, slope = velocity. |
| **Average speed** | Total distance ÷ time. A scalar — no direction. Velocity adds the sign. |

## SECTION B KEY EQUATIONS + UNITS

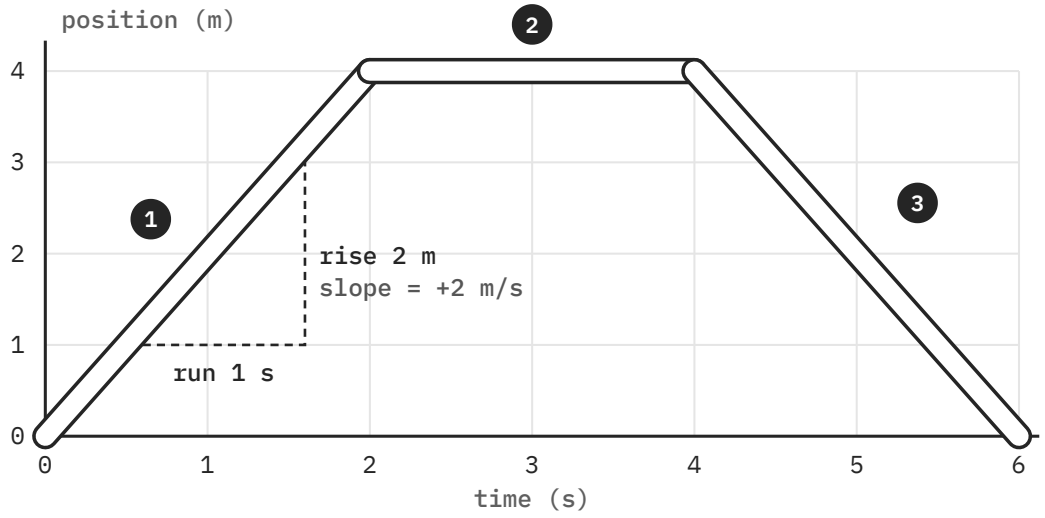
$$v = \Delta x \div \Delta t$$

speed = distance ÷ time

| | |
|---|---|
| v | m/s (vector: has a sign) |
| convert | 1 m/s = 3.6 km/h |

## SECTION C COLOR IT · READ A GRAPH AS A MOTION STORY

*Height = position. Slope = velocity.*




☐ **1 · PURPLE** Rising: moving +, v > 0

☐ **2 · PURPLE** Flat: stopped, v = 0

☐ **3 · PURPLE** Falling: moving –, v < 0

## SECTION D WORKED EXAMPLE

A cart goes from 8 m to 2 m in 3 s. Right is positive. Find its velocity.

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| $x_i$ = 8 m<br>$x_f$ = 2 m<br>$\Delta t$ = 3 s | $v = \Delta x \div \Delta t$ | $\Delta x$ = 2 − 8 = −6 m<br>v = −6 m ÷ 3 s | **−2 m/s**<br>2 m/s to the left |

## SECTION E THE 2026-XJ THREAD

DAY 3 OF 22 · NEW TOOL: VELOCITY

Two position readings and the time between them give XJ's velocity: the slope of its position–time graph. That tells us how fast it closes in, and from which side.

*Unit question: How fast is it moving, and where will it arrive?*

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Read motion from the slope

Velocity, speed, and position-time graphs

A position tells us where an object is. To describe its motion, we need to know how that position changes over time. Moving six meters in three seconds and moving six meters in thirty seconds are different motions. Velocity combines the position change, the elapsed time, and the direction.

Use the average-velocity equation in **Section B**: $\boldsymbol{v = \Delta x \div \Delta t}$. The units are meters per second, or m/s. Average speed uses total distance divided by elapsed time. Speed has no direction and cannot be negative. When a trip includes a return, average speed and the size of average velocity can be different because distance and displacement are different.

A position-time graph places time on the horizontal axis and position on the vertical axis. Its height tells us position. Its slope tells us velocity. Slope means rise divided by run: the change in position divided by the change in time. A rising line means motion in the positive direction. A falling line means motion in the negative direction.

Follow the rising, flat, and falling parts of the graph in **Section C**. From **0 s to 2 s**, position rises from **0 m to 4 m**. The slope is **4 m ÷ 2 s = +2 m/s**. From 2 s to 4 s, position stays at 4 m. The flat line has zero slope, so the object is stopped. From 4 s to 6 s, it returns to 0 m, giving **–4 m ÷ 2 s = –2 m/s**.

The return slope is negative, but its size matches the first slope. The object moves at the same speed in the opposite direction. A steeper position-time line would mean a greater speed. The graph is not a picture of a hill or road. Its shape records how position changes as time passes. The purple marks connect those slopes with velocity.

In **Section D**, the cart goes from **8 m to 2 m in 3 s**. Right is positive. First find displacement: $\boldsymbol{\Delta x}$ = **2 m – 8 m = –6 m**. Then divide by time: $\boldsymbol{v}$ = **–6 m ÷ 3 s = –2 m/s**. That means an average velocity of two meters per second to the left. The minus sign gives direction, not a negative speed.

The notes also give a useful unit conversion: **1 m/s = 3.6 km/h**. A speed of 2 m/s is therefore **(2)(3.6) = 7.2 km/h**. Changing units changes the number used to describe the motion, not the motion itself.

For our fictional asteroid, 2026-XJ, measured positions and their times let us estimate velocity. In the lab, the same idea applies to your cart: use the actual graph and its labeled axes. I want the slope calculation and your description of direction to tell the same story.

# Velocity–time graphs and vectors

**LEARNING TARGET · R2** I can read a velocity–time graph and split a velocity arrow into two components.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Velocity–time graph** | The vertical axis shows *velocity*, not position. Read the label first. |
| **Flat v–t line** | Steady velocity — not necessarily stopped. |
| **Area under the line** | Height × width = displacement. Area below zero is negative displacement. |
| **Components** | The separate east and north changes that make up one diagonal displacement. |

## SECTION B KEY EQUATIONS + UNITS

$$\Delta x = v \times t \quad \text{(area)}$$

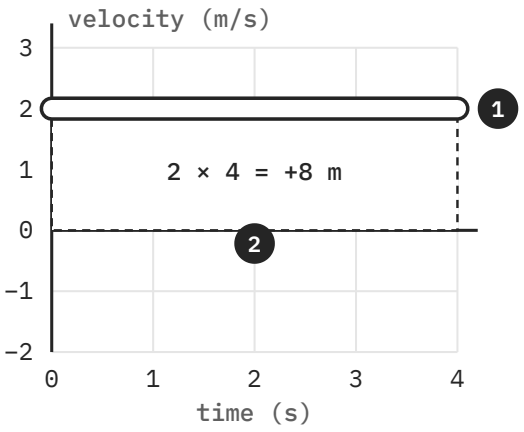
$$v_x = v \cos \theta$$

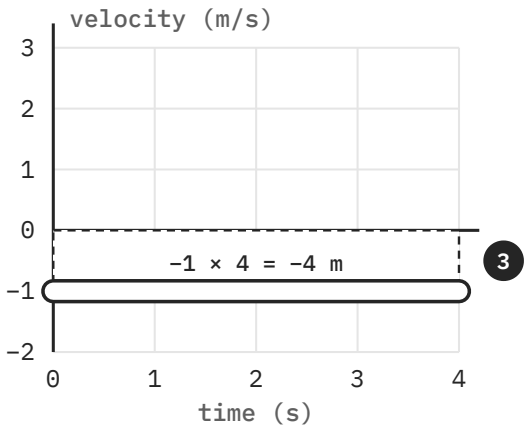
$$v_y = v \sin \theta$$

| | |
|---|---|
| $\theta$ | measured from the +x axis |
| cos / sin | part next to / opposite $\theta$ |

## SECTION C COLOR IT · AREA COUNTS THE CHANGE

*Two velocity graphs. Right is +.*




☐ **1 · PURPLE** Velocity lines

☐ **2 · BLUE** Area above zero: + displacement

☐ **3 · BLUE** Area below zero: stripes, − displacement

## SECTION D WORKED EXAMPLE

Velocity is +3 m/s for 2 s. Find the displacement and name its direction.

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| v = +3 m/s<br>t = 2 s | $\Delta x = v \times t$ | $\Delta x = (3\ \text{m/s})(2\ \text{s})$ | **+6 m**<br>6 m to the right |

## SECTION E THE 2026-XJ THREAD

DAY 4 OF 22 · NEW TOOL: V–T GRAPHS + COMPONENTS

A velocity–time graph shows whether XJ's speed is holding steady. Splitting its velocity arrow into parts shows how much is aimed at Earth and how much carries it sideways.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Read the axes before you read the line

Velocity-time graphs, signed area, and components

A flat line does not always mean an object is stopped. On yesterday's position-time graph, a flat line meant position stayed the same. On today's velocity-time graph, it means velocity stays the same. I want the axis labels to be the first thing you read. They tell you what the line's height actually represents.

If the line stays at **+2 m/s**, the object keeps moving in the positive direction at two meters per second. If it stays at **−1 m/s**, the object keeps moving in the negative direction at one meter per second. Only a line at zero velocity describes an object remaining stopped. A negative velocity is a direction, not automatically slowing motion.

The area between a velocity line and the time axis gives displacement. For a constant velocity, that area is a rectangle. Its height is velocity and its width is elapsed time, so $\Delta x = v \times t$. Multiplying meters per second by seconds leaves meters, the unit for a change in position.

Find the time axis on both graphs in **Section C**. The first graph shows **+2 m/s for 4 s**. Its area gives $\Delta x = (2\ \text{m/s})(4\ \text{s}) = +8\ \text{m}$. The second shows **−1 m/s for 4 s**, giving $\Delta x = (-1\ \text{m/s})(4\ \text{s}) = -4\ \text{m}$. Area above zero adds positive displacement. Area below zero adds negative displacement. The blue regions represent those position changes, while the velocity lines are purple.

The example in **Section D** uses **+3 m/s for 2 s**. Apply the same rule: $\Delta x = (3\ \text{m/s})(2\ \text{s}) = +6\ \text{m}$. With right chosen as positive, the cart ends six meters right of where it began. This is displacement, not its final coordinate. To find the final coordinate, we would also need the starting position.

A velocity arrow can point diagonally rather than along one axis. We then describe it with two components: a horizontal part and a vertical part. For an angle $\theta$ measured from the positive horizontal axis, $v_x = v\cos\theta$ and $v_y = v\sin\theta$. Cosine finds the part next to that angle; sine finds the opposite part. Both components describe the same motion happening at the same time.

For example, a 10 m/s velocity at 30° above that axis has components $v_x = 10\cos 30° \approx 8.66\ \text{m/s}$ and $v_y = 10\sin 30° = 5\ \text{m/s}$. Use degree mode and check the arrow's direction before assigning signs.

For our fictional asteroid, 2026-XJ, the velocity graph shows how motion changes with time. Components show how much motion is toward Earth and how much is sideways. We need both pieces to predict a path, rather than assuming that all of its speed carries it directly toward us.

# Predicting position and uncertainty

**LEARNING TARGET · R3** I can use velocity and time to predict where it will be, including the possible error.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Starting position** | Where the object is at t = 0. It is often not zero. |
| **Change** | Velocity × time: how far the position shifts. |
| **Final position** | Start + change. |
| **Prediction range** | A ± bound. "10 ± 2 m" means 8 m to 12 m. More time usually means a wider range. |

## SECTION B KEY EQUATIONS + UNITS

$$x_f = x_0 + v\,t$$

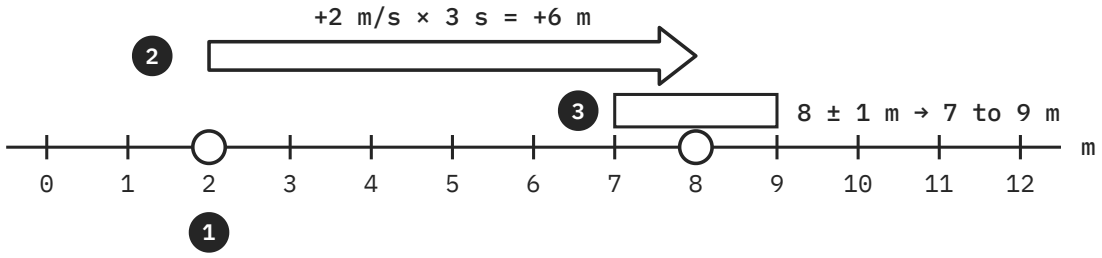
largest error = start error + (velocity error × time)

**units first** 2 h = 7,200 s

**2-D** also $y_f = y_0 + v_y t$

## SECTION C COLOR IT · START, CHANGE, RANGE

*Start 2 m, +2 m/s for 3 s.*




☐ **1 · BLUE** Starting position (dot)

☐ **2 · BLUE** The change, +6 m (arrow)

☐ **3 · BLUE** The range, 7 to 9 m (light shade)

## SECTION D WORKED EXAMPLE

A cart starts at 8 m and moves at −2 m/s for 3 s. Where is it at the end?

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| $x_i$ = 8 m<br>v = −2 m/s<br>t = 3 s | $x_f = x_i + v \times t$ | change = (−2)(3) = −6 m<br>$x_f$ = 8 − 6 | **2 m**<br>moved 6 m left |

## SECTION E THE 2026-XJ THREAD

DAY 5 OF 22 · NEW TOOL: FORECAST + UNCERTAINTY

With a starting position and a velocity, we can forecast where XJ will be. But every measurement has error, so a responsible forecast is a range, not a single point.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Start with the starting position

Steady-velocity predictions and their possible range

Imagine a cart already sitting at the eight-meter mark. If it moves six meters left, it does not finish at negative six meters. Negative six meters describes its change in position. We still have to combine that change with where the cart started. Today I want you to keep those two steps visible.

For steady velocity, the position change is velocity times elapsed time. Final position is starting position plus that change: $x_f = x_0 + vt$. The subscript zero identifies the position when our chosen time interval begins. The starting position can be positive, negative, or zero. We should copy it from the situation, not assume it.

Find the starting dot and the displacement arrow in **Section C**. The trip begins at **2 m** and shows motion at **+2 m/s for 3 s**. First calculate **change = (2 m/s)(3 s) = +6 m**. Then $x_f$ **= 2 m + 6 m = 8 m**. The dot marks the starting position, and the blue arrow marks the change. The answer is a coordinate measured from the same zero used at the start.

In **Section D**, the cart starts at **8 m** and moves at **−2 m/s for 3 s**. The negative velocity means motion left when right is positive. Its change is **(−2 m/s)(3 s) = −6 m**. Add that signed change to the start: $x_f$ **= 8 m + (−6 m) = 2 m**. The final position remains positive even though the cart moved in the negative direction.

Measurements also have limits. A prediction written as **8 ± 1 m** means a central value of 8 m with an allowed range from **7 m to 9 m**. The plus-or-minus amount extends to each side of the center. It is not the full width. Here the full interval is two meters wide, as the shaded region shows.

For steady velocity and an exact elapsed time, we can use a conservative bound: **largest position error = start error + velocity error × time**. If the starting error is 0.5 m and velocity error is 0.1 m/s, after three seconds the bound is **0.5 m + (0.1 m/s)(3 s) = 0.8 m**. More elapsed time lets the velocity uncertainty build into more position uncertainty. This range does not state an impact probability.

Check units before multiplying. If velocity is in meters per second, time needs to be in seconds: **2 h = 7,200 s**. In two dimensions, apply the same start-plus-change idea separately to the vertical coordinate: $y_f = y_0 + v_y t$.

For our fictional asteroid, 2026-XJ, a responsible forecast includes the future time, predicted location, range, and steady-motion assumption. If its velocity changes, we need to revise the model rather than trust a precise-looking answer from the wrong equation.

# Vector addition and decomposition

**LEARNING TARGET · S2** I can join arrows tip to tail and split an arrow into parts using sine and cosine.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Vector** | A quantity with size and direction, drawn as an arrow. |
| **Tip-to-tail** | Start each arrow where the last one ends. Do not turn or stretch it. |
| **Component** | The east–west or north–south part of a vector. |
| **Resultant** | The overall arrow from the first start to the final finish. |

## SECTION B KEY EQUATIONS + SIGNS

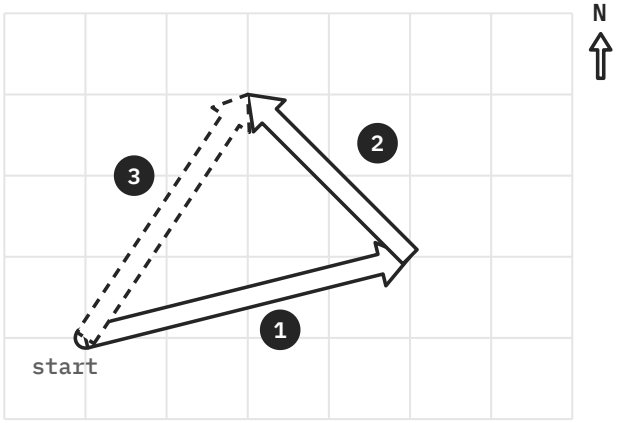
$$R_x = A_x + B_x \cdot R_y = A_y + B_y$$
$$|R| = \sqrt{(R_x^2 + R_y^2)}$$
$$\theta = \tan^{-1}(R_y \div R_x)$$

parts: V cos θ (next to θ) · V sin θ (opposite)

\+ east, north
– west, south

## SECTION C COLOR IT · TIP TO TAIL

*Each square = 1 m. No trig needed.*




A: 4 E, 1 N

B: 2 W, 2 N

sum: 2 E, 3 N

E–W: 4 – 2 = 2 E
N–S: 1 + 2 = 3 N

☐ **1 · BLUE** Arrow A

☐ **2 · BLUE** Arrow B

☐ **3 · BLUE** Resultant (dashed)

## SECTION D WORKED EXAMPLE

A 20 N pull points 30° above the +x axis. Split it into its x and y parts.

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| F = 20 N<br>θ = 30° from +x | $F_x = F \cos \theta$<br>$F_y = F \sin \theta$ | 20 cos 30°<br>20 sin 30° | **17.3 N, 10.0 N**<br>parts of one force, not two new forces |

## SECTION E THE 2026-XJ THREAD

DAY 6 OF 22 · NEW TOOL: VECTOR ADDITION

XJ's motion has parts in more than one direction. Adding them tip to tail gives its true path; splitting them shows the part that actually closes on Earth.

*Unit question: How fast is it moving, and where will it arrive?*

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Keep direction attached to every arrow

Tip-to-tail addition and splitting vectors into parts

An arrow gives us two pieces of information: size and direction. Together they describe a vector. Displacement, velocity, and force are vectors, so their directions must stay attached when we combine them. Adding only the arrow lengths can give the wrong result when the arrows point different ways.

Follow the arrows tip to tail in **Section C**. They show displacement A going **4 m east and 1 m north**, followed by displacement B going **2 m west and 2 m north**. To add them visually, place the tail of B at the tip of A. Keep each arrow's length and direction unchanged. The resultant is the arrow from the first starting point to the final endpoint.

We can check that picture by adding components. Take east and north as positive. Horizontally, $\boldsymbol{R_x} = \mathbf{4\ m - 2\ m = 2\ m}$ east. Vertically, $\boldsymbol{R_y} = \mathbf{1\ m + 2\ m = 3\ m}$ north. In symbols, $\boldsymbol{R_x} = \boldsymbol{A_x} + \boldsymbol{B_x}$ and $\boldsymbol{R_y} = \boldsymbol{A_y} + \boldsymbol{B_y}$. Add horizontal parts together and vertical parts together.

Those perpendicular parts form a right triangle. The resultant's size is $|\boldsymbol{R}| = \sqrt{[(\mathbf{2\ m})^2 + (\mathbf{3\ m})^2]} \approx \mathbf{3.61\ m}$. Its angle from the positive horizontal direction is $\boldsymbol{\theta} = \tan^{-1}(\mathbf{3 \div 2}) \approx \mathbf{56.3}°$, toward north from east. Check both signs before interpreting an angle. The arrow here must point right and up because both components are positive.

We can also work backward, splitting one diagonal arrow into its parts. The force example in **Section D** gives a **20 N force at 30° above the positive horizontal axis**. The horizontal part is next to the angle, so use cosine: $\boldsymbol{F_x} = \boldsymbol{F}\cos\boldsymbol{\theta} = \mathbf{(20\ N)\cos 30°} \approx \mathbf{17.3\ N}$. The vertical part is opposite, so $\boldsymbol{F_y} = \boldsymbol{F}\sin\boldsymbol{\theta} = \mathbf{(20\ N)\sin 30°} = \mathbf{10.0\ N}$.

Both force components are positive in this example. Use degree mode, and name the axis from which the angle is measured. If the angle begins at another axis, identify the adjacent and opposite parts again. I want the sketch to guide the sine-or-cosine choice.

The 17.3 N and 10.0 N values describe parts of one 20 N force. They are not two new forces to add on top of it. Similarly, the blue displacement components describe one overall change in position. Components organize the calculation without creating extra motion or extra interactions.

For our fictional asteroid, 2026-XJ, motion has parts in more than one direction. Adding vectors keeps those directions in the prediction. Splitting a velocity arrow helps us distinguish motion toward Earth from motion across our line of sight.

# Acceleration

**LEARNING TARGET · R4** I can use velocity–time data to find how fast velocity changes.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Acceleration** | The rate velocity changes: how much velocity is gained or lost each second. |
| **Speeding up** | v and a have the same sign. Gaps between equal-time dots grow. |
| **Slowing down** | v and a have opposite signs. Braking is acceleration too. |
| **Turning** | A direction change is a velocity change, even at steady speed. |

## SECTION B KEY EQUATIONS + UNITS

$$a = \Delta v \div \Delta t$$

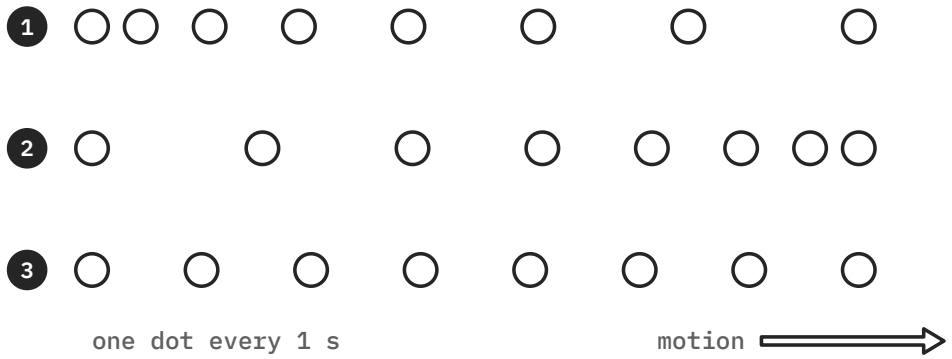
$$v = v_0 + a\,t$$

a m/s² ("m/s, every second")

g ≈ 9.8 m/s² (free fall)

## SECTION C COLOR IT · READ THE GAPS

*Equal time between dots.*




☐ **1 · GREEN** Speeding up: gaps grow

☐ **2 · RED** Slowing down: gaps shrink

☐ **3 · YELLOW** Steady: a = 0

## SECTION D WORKED EXAMPLE

A cart's velocity changes from 0 to 8 m/s in 4 s. Find its acceleration.

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| $v_i = 0$ m/s<br>$v_f = 8$ m/s<br>$\Delta t = 4$ s | $a = \Delta v \div \Delta t$ | $\Delta v = 8 - 0 = 8$ m/s<br>$a = 8 \div 4$ | **2 m/s²**<br>gains 2 m/s each second |

## SECTION E THE 2026-XJ THREAD

DAY 7 OF 22 · NEW TOOL: ACCELERATION

If XJ's velocity readings change from one night to the next, it is accelerating, and a forecast that assumes steady velocity will miss.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Acceleration means velocity is changing

Reading dot patterns and calculating change each second

We often use the word accelerate to mean speed up, but in physics it has a wider meaning. Acceleration is a change in velocity over time. Speeding up is one example. Slowing down and turning are examples too, because velocity includes both speed and direction. I want us to describe the change before deciding what the acceleration means.

The calculation is $a = \Delta v \div \Delta t$. First subtract starting velocity from final velocity. Then divide by elapsed time. The unit m/s² means meters per second, each second. An acceleration of 2 m/s² means velocity changes by two meters per second during each second if the acceleration stays constant.

The cart in **Section D** starts at **0 m/s** and reaches **8 m/s in 4 s**. The velocity change is $\Delta v = 8\ \text{m/s} - 0\ \text{m/s} = 8\ \text{m/s}$. Divide by the time: $a = 8\ \text{m/s} \div 4\ \text{s} = 2\ \text{m/s}^2$. That is the average acceleration over the interval. With steady acceleration, the velocities at successive whole seconds would be 0, 2, 4, 6, and 8 m/s.

The second equation in **Section B** gives $v = v_0 + at$. This says final velocity equals starting velocity plus the change built up over time. For our steady-acceleration example, $v = 0\ \text{m/s} + (2\ \text{m/s}^2)(4\ \text{s}) = 8\ \text{m/s}$. The two equations describe the same change from different starting information.

Compare the gaps between dots in the three rows of **Section C**. They mark position at equal time intervals. Growing gaps mean the object travels farther during each interval, so it speeds up. Shrinking gaps mean it travels less during each interval, so it slows down. Equal gaps along a straight path mean steady velocity and zero acceleration. Green, red, and yellow mark those three patterns on the notes page.

A negative acceleration does not always mean slowing down. Along a straight line, compare the signs of velocity and acceleration. When they have the same sign, speed grows. When they have opposite signs, speed decreases until a possible turn. A cart moving right can slow under leftward acceleration; a cart already moving left can speed up under that same acceleration.

Turning matters too. An object going around a curve at steady speed still changes its direction, so its velocity changes and it accelerates. Near Earth's surface, free fall is another example: ignoring air resistance, gravity produces about **9.8 m/s² downward**. The sign depends on which direction we choose as positive.

For our fictional asteroid, 2026-XJ, changing velocity readings tell us that a steady-velocity forecast may need revision. The important question is how much velocity changes, over what time, and in which direction. That information lets us begin explaining a changing path instead of simply extending an old pattern.

# Constant-acceleration problem-solving

**LEARNING TARGET · R5** I can choose and use an equation for motion with steady acceleration.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Steady velocity** | Equal gaps; velocity does not change. Use v × t. |
| **Constant acceleration** | Velocity changes by the same amount each second. |
| **Average velocity** | (start + final) ÷ 2 — only when acceleration is constant. |
| **Method** | A choice with a reason: "I chose ___ because the velocity ___." |

## SECTION B KEY EQUATIONS + UNITS

$$v = v_0 + a\,t$$

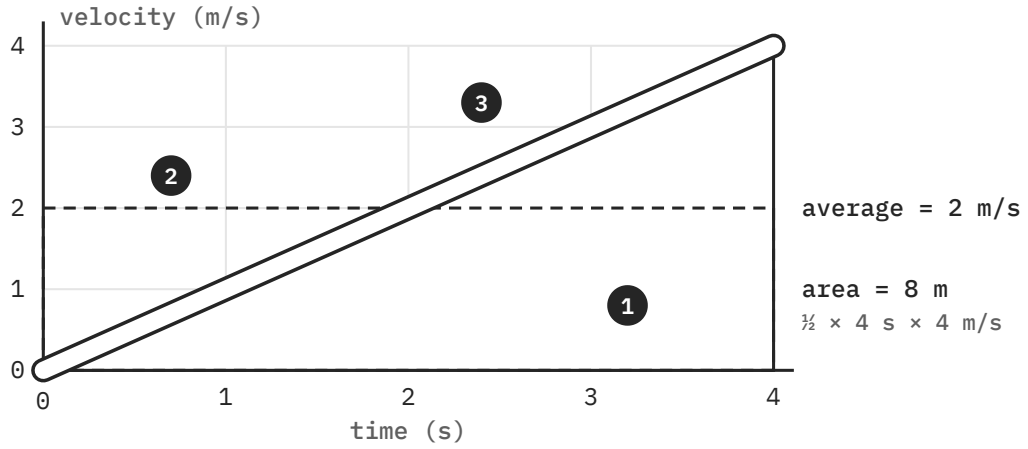
$$\Delta x = v_0 t + \tfrac{1}{2} a t^2$$

$$v^2 = v_0^2 + 2a\Delta x \quad \text{(no t)}$$

**1 · 2** list given · list asked
**3 · 4** choose · check units + sign

## SECTION C COLOR IT · TRIANGLE OR AVERAGE: SAME AREA

*0 → 4 m/s in 4 s, constant a.*




☐ **1 · BLUE** Triangle area = displacement

☐ **2 · PURPLE** Average-velocity box (dashed)

☐ **3 · PURPLE** Velocity line

## SECTION D WORKED EXAMPLE

A cart speeds up steadily from 0 to 8 m/s in 4 s. Find its displacement.

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| $v_i = 0$, $v_f = 8$ m/s<br>t = 4 s<br>constant a | $\Delta x = \tfrac{1}{2}(v_i + v_f) \times t$ | average = (0 + 8) ÷ 2 = 4 m/s<br>$\Delta x = 4 \times 4$ | **+16 m**<br>average works: a is constant |

## SECTION E THE 2026-XJ THREAD

DAY 8 OF 22 · NEW TOOL: CHOOSING A MODEL

Steady velocity or steady acceleration? Choosing the right model for XJ is the difference between a forecast that holds and one that drifts.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Choose a motion model before an equation

Steady acceleration and displacement from average velocity

A cart traveling steadily at 8 m/s for four seconds covers a different distance from a cart that starts at rest and only reaches 8 m/s at the end. Using the final velocity for the whole interval would pretend the second cart was already moving that fast at the start. Today I want the equation to match the actual motion.

Steady velocity means velocity does not change. In that case, $\Delta x = vt$. Constant acceleration means velocity changes by the same amount each second. For constant acceleration, the average velocity is $v_{avg} = (v_i + v_f) \div 2$. Averaging only the starting and ending velocities works because the change between them is steady.

The cart in **Section D** speeds up steadily from **0 m/s to 8 m/s in 4 s**. First find the average: $v_{avg} = (0 + 8)\ \text{m/s} \div 2 = 4\ \text{m/s}$. Then multiply by elapsed time: $\Delta x = (4\ \text{m/s})(4\ \text{s}) = +16\ \text{m}$. The cart's position changes sixteen meters in the positive direction.

We can combine those steps into $\Delta x = \frac{1}{2}(v_i + v_f)t$. The one-half simply tells us to divide the sum of the endpoint velocities by two. The result is displacement. If the object reversed direction during another problem, we would need to distinguish that signed change from the total distance traveled.

Trace the triangle and the dashed rectangle in **Section C**. This graph uses a smaller final velocity: **0 to 4 m/s in 4 s**. The straight rising line shows constant acceleration. Its triangular area is $\frac{1}{2}(4\ \text{s})(4\ \text{m/s}) = 8\ \text{m}$. The average-velocity rectangle has height 2 m/s and the same width, giving $(2\ \text{m/s})(4\ \text{s}) = 8\ \text{m}$. Different shapes give the same area and therefore the same displacement.

The constant-acceleration equations in **Section B** connect other sets of known values. $v = v_0 + at$ finds velocity from its starting value, acceleration, and time. $\Delta x = v_0 t + \frac{1}{2}at^2$ finds displacement from those quantities. $v^2 = v_0^2 + 2a\Delta x$ is useful when time is not given or needed.

Start by listing the given values with units and naming what is asked. Then choose a relationship that uses that information and fits the model. Finish by checking the units and signs. A calculation for displacement should end in meters, and its sign should agree with the motion. Do not assume constant acceleration just because an equation is available.

For our fictional asteroid, 2026-XJ, choosing steady velocity or steady acceleration is a decision about the evidence. I want the reason for that choice to travel with the predicted position. A correct substitution cannot rescue a model that does not fit the motion.

# Newton's first law and inertia

**LEARNING TARGET · R6** I can explain Newton's 1st Law in my own words — and why 2026-XJ keeps moving.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Force** | A push or pull from an interaction with another object. |
| **Net force** | The combined effect of all forces on one object. |
| **Newton's 1st law** | *1st law:* at rest stays at rest; moving keeps its velocity — unless $\Sigma F \neq 0$. |
| **Inertia** | An object keeps its velocity unless a net force changes it. More mass, more inertia. |

## SECTION B KEY IDEA + UNITS

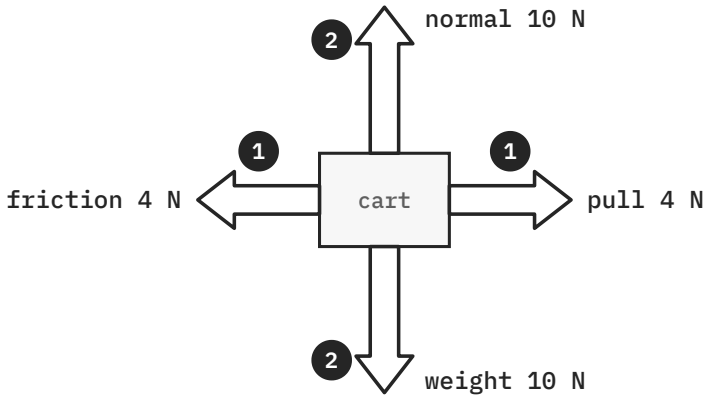
$$\Sigma F = 0$$
$$\rightarrow a = 0$$
$$\rightarrow \text{velocity stays the same}$$

in an inertial frame (not speeding up or turning)

`force` newtons (N)
`mass` kilograms (kg)

## SECTION C COLOR IT · BALANCED, STILL MOVING

*A cart rolls right; forces balance.*




- ☐ **1 · ORANGE** Horizontal forces (balanced)
- ☐ **2 · ORANGE** Vertical forces (balanced)
- ☐ **3 · PURPLE** Velocity arrow (not a force)

## SECTION D WORKED EXAMPLE

A cart rolls right, then slows down. What does that tell you about its net horizontal force?

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| moving right<br>speed decreasing | $a \neq 0 \rightarrow$ net force $\neq 0$ | velocity changes →<br>there is a net force; it opposes the motion | **Net force left**<br>opposite the motion |

## SECTION E THE 2026-XJ THREAD

DAY 9 OF 22 · NEW TOOL: INERTIA

Out in space, nothing pushes XJ along. It keeps moving because of inertia, so no force is needed to explain its motion — only to explain a change in it.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Moving does not require an unbalanced force

Newton's first law, balanced forces, and inertia

A cart moving across a rough floor slows down after we stop pushing. It is easy to conclude that every moving object needs a forward force to keep going. But the cart is still interacting with the floor and air. We need to identify those forces before deciding what motion would look like without them.

A force is a push or pull from an interaction. The net force is the combined effect of all the forces on one object. Newton's first law says an object keeps its velocity unless a nonzero net force changes it. If the object is at rest, it stays at rest. If it is already moving, it continues at constant speed in a straight line.

This rule applies in an inertial frame: a reference frame that is not itself accelerating or turning. We treat the classroom as such a frame for these examples. The key relationship is $\boldsymbol{\Sigma F = 0}$, which gives $\boldsymbol{a = 0}$ and unchanged velocity. The symbol Σ means add the forces, including their directions.

Compare the two horizontal arrows on the moving cart in **Section C**. A **4 N pull right** balances **4 N of friction left**. A **10 N normal force up** balances **10 N of weight down**. Each direction balances, so the cart has zero acceleration. It can still keep rolling right. The purple velocity arrow describes that motion and is not another force to add to the orange force arrows.

Inertia is an object's resistance to a change in velocity. More mass means more inertia. It is not a hidden forward force. A moving object does not need an additional “inertia arrow” in its force diagram. We use forces to explain changes in motion, rather than inventing a force merely because something is moving.

Now compare that cart with the one in **Section D**: a cart rolls right and slows down. Its velocity is changing, so its acceleration is not zero. The change points left, opposite the rightward motion. Therefore its net horizontal force must point left. Friction could cause that imbalance, but the observed motion alone tells us the net direction, not the exact size of each individual force.

If someone keeps pulling the cart right with a force equal to the backward friction, the forces balance and the cart can move steadily. If the pull becomes smaller than friction, the cart slows while moving right. This is why “there is a force” and “there is a net force” are different statements. Several forces can act and still add to zero.

For our fictional asteroid, 2026-XJ, there is no need for a forward push simply to maintain its motion. Gravity can still change its velocity and bend its path. I want us to distinguish motion that continues because of inertia from motion that changes because of a net external force.

# Forces: what pushes and pulls

**LEARNING TARGET · K2** I can name the forces in a scenario: gravity, normal, friction, applied, tension.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Interaction** | Every force has a source (what pushes or pulls) and a receiver. |
| **Contact force** | Normal (surface, ⊥), friction (along surface, opposes slip), tension (rope), applied (a push). |
| **Non-contact force** | Gravity acts without touching. |
| **Mass vs. weight** | Mass is amount of matter (kg). Weight is Earth's gravitational pull (N). |

## SECTION B KEY EQUATION + UNITS

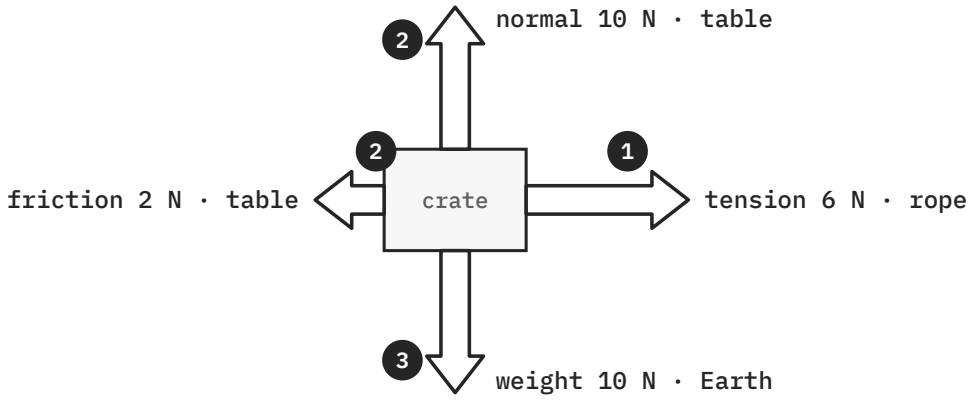
$$F_g = m\ g$$

g = 9.8 N/kg (≈ 10 to estimate)

m mass, kilograms (kg)
W weight, newtons (N)

## SECTION C COLOR IT · NAME EVERY SOURCE

*The ___ exerts a ___ force on the crate.*




☐ **1 · ORANGE** Force from the rope

☐ **2 · ORANGE** Forces from the table

☐ **3 · ORANGE** Force from Earth

## SECTION D WORKED EXAMPLE

A 3 kg object rests on a level table. Find its weight and name the force that supports it.

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| m = 3 kg<br>g ≈ 10 N/kg | W = m × g | W = 3 × 10 = 30 N down<br>table pushes up equally | **30 N**<br>normal force 30 N up |

## SECTION E THE 2026-XJ THREAD

DAY 10 OF 22 · NEW TOOL: NAMING FORCES

If XJ's path ever bends or speeds up, something is pulling on it. Naming the source of every force (out there, mostly gravity) is how analysts explain a change.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Name what is doing the pushing or pulling

Force sources, contact forces, mass, and weight

A force arrow should tell us about an interaction, not just point in a direction that looks useful. I want you to be able to finish this sentence for every arrow: “The ___ exerts a ___ force on the object.” The first blank identifies the source. The object receiving the force is the receiver.

Some forces require contact. A normal force is a surface's push perpendicular to the surface. Friction acts along the contact surface and opposes slipping or the tendency to slip. Tension is a taut rope's pull along the rope. An applied force can be a person's push or pull. These names describe how objects interact.

Gravity can act without contact. Near Earth's surface, Earth pulls objects downward. We call the size of that gravitational force weight. Mass describes how much matter an object has and is measured in kilograms. Weight is a force and is measured in newtons. A three-kilogram object does not have a weight of three kilograms; those units describe its mass.

Use the weight equation in **Section B**: $\boldsymbol{W = mg}$, also written using gravitational force as $\boldsymbol{F_g = mg}$. Near Earth, $\boldsymbol{g \approx 9.8\ N/kg}$. That means each kilogram has about 9.8 N of gravitational force acting on it. For estimates and today's worked example, the notes round this to **10 N/kg**.

The worked example places a **3 kg object** at rest on a level table. Its estimated weight is $\boldsymbol{W = (3\ kg)(10\ N/kg) = 30\ N}$ downward. The table supplies an upward normal force. Because the object remains at rest and there are no other vertical forces in this example, the normal force must also be **30 N**. Equal upward and downward forces give zero vertical acceleration.

Name the source of each of the four force arrows on the crate in **Section C**. The rope pulls right with 6 N of tension. The table supplies 2 N of backward friction and 10 N of upward normal force. Earth supplies 10 N of downward weight. One source, the table, can supply two different contact forces. The vertical forces balance, while the horizontal forces have a net effect to the right.

Normal force is not automatically equal to weight. That equality followed from the level-table situation with no other vertical pull and no vertical acceleration. A ramp or another supporting force can change it. Similarly, a velocity arrow is not a force source. “It is moving” cannot fill in the first blank of our interaction sentence.

For our fictional asteroid, 2026-XJ, naming sources helps us decide which forces belong in its model. Gravity from other bodies can change its motion without contact. A rope or table force belongs only if such an interaction exists. A complete, realistic force list is the starting point for explaining a changing path.

# Newton's second law: vary mass

**LEARNING TARGET · S3** I can measure how acceleration changes with mass at constant force.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Fair test** | Change one thing at a time. |
| **Changed variable** | Total system mass: cart + added masses + hanger. |
| **Controls** | Hanging mass, track, start point, release (no push). |
| **Evidence** | The slope of each velocity–time graph is the acceleration. |

## SECTION B KEY EQUATION + PATTERN

$$a \propto 1 \div m$$

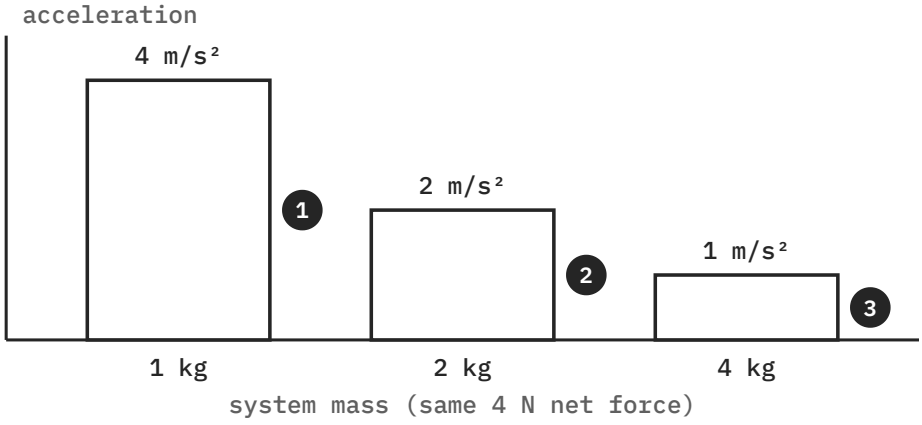
$$m \times a \approx \text{constant}$$

$$a = \Delta v \div \Delta t \text{ (slope)}$$

mass kilograms (kg)
a m/s²

## SECTION C COLOR IT · DOUBLE THE MASS

*Illustrative values — not your data.*




☐ **1 · GREEN** 1 kg system (dark)
☐ **2 · GREEN** 2 kg system (medium)
☐ **3 · GREEN** 4 kg system (light)

## SECTION D WORKED EXAMPLE

In one trial, velocity rises from 0 to 4 m/s in 2 s. Find the acceleration.

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| $v_i = 0$ m/s<br>$v_f = 4$ m/s<br>$\Delta t = 2$ s | $a = \Delta v \div \Delta t$ | $\Delta v = 4 - 0 = 4$ m/s<br>$a = 4 \div 2$ | **2 m/s²**<br>use your own measured graph |

## SECTION E THE 2026-XJ THREAD

DAY 11 OF 22 · NEW TOOL: MASS

A massive object is hard to nudge. The bigger XJ's mass, the smaller the acceleration any push or pull gives it, which matters for any plan to deflect it.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Change the mass and watch the acceleration

A fair test with the same net force

Imagine pushing an empty cart and then the same cart loaded with more mass. With the same net force, the loaded cart changes velocity more slowly. Today's lab tests that relationship. I want the evidence to come from measured motion, with the force conditions kept as consistent as possible.

A fair test changes one intended quantity at a time. Here that quantity is the total moving mass. In our cart-and-hanger system, count the cart, added masses, and hanger if all of them move as part of the system. Leaving out part of the moving mass would make the comparison incomplete.

Keep the hanging mass, track, starting point, and release method consistent as the notes describe. Releasing without an extra push matters because a different starting motion could change the trial. Holding the hanging mass steady keeps the driving pull similar. Friction and other resistance still need attention because the relationship we want concerns the net force, not merely the hanging weight.

Use the two velocities and elapsed time in **Section D**. We find acceleration from the slope of each velocity-time graph: $a = \Delta v \div \Delta t$. In the worked example, velocity rises from **0 m/s to 4 m/s in 2 s**. First find $\Delta v = 4\ \text{m/s} - 0\ \text{m/s} = 4\ \text{m/s}$. Then $a = 4\ \text{m/s} \div 2\ \text{s} = 2\ \text{m/s}^2$. Your experiment uses your own measured graph, not these example readings.

Compare the three masses in **Section C**. The picture gives illustrative values for a fixed **4 N net force**. A 1 kg system accelerates at 4 m/s², a 2 kg system at 2 m/s², and a 4 kg system at 1 m/s². Doubling mass halves acceleration when the net force stays the same. This is an inverse relationship: as mass increases, acceleration decreases in that particular way.

Another check is the product of mass and acceleration. The examples give $(1\ \text{kg})(4\ \text{m/s}^2) = 4\ \text{N}$, $(2\ \text{kg})(2\ \text{m/s}^2) = 4\ \text{N}$, and $(4\ \text{kg})(1\ \text{m/s}^2) = 4\ \text{N}$. Each product is the same because the net force is the same. Real measurements may not match perfectly, so look for a pattern while considering measurement uncertainty and changes in resistance.

I want a conclusion to name both parts of the comparison: increasing total mass reduces acceleration under the same net force. Saying only “more mass means less acceleration” leaves out the condition. If the force changes too, we cannot predict the result from mass alone. The green acceleration arrows in the diagram show the relative sizes in the illustrated speeding-up cases.

For our fictional asteroid, 2026-XJ, mass matters when considering a proposed push. The same net push has less effect on a larger mass. But gravity's force can also change with mass, so we should not use today's fixed-force comparison to claim that all heavier objects fall more slowly. We must keep track of what is actually being held constant.

# Newton's second law: vary force

**LEARNING TARGET · S4** I can measure how acceleration changes with force at constant mass — and F = ma emerges.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Driving force** | The hanging weight that pulls the cart system. |
| **Total moving mass** | Held constant: move masses from cart to hanger; never add new ones. |
| **Proportional** | Double the net force → double the acceleration. |
| **Friction offset** | Friction lowers the net force, so real data can sit below the ideal line. |

## SECTION B KEY EQUATIONS + UNITS

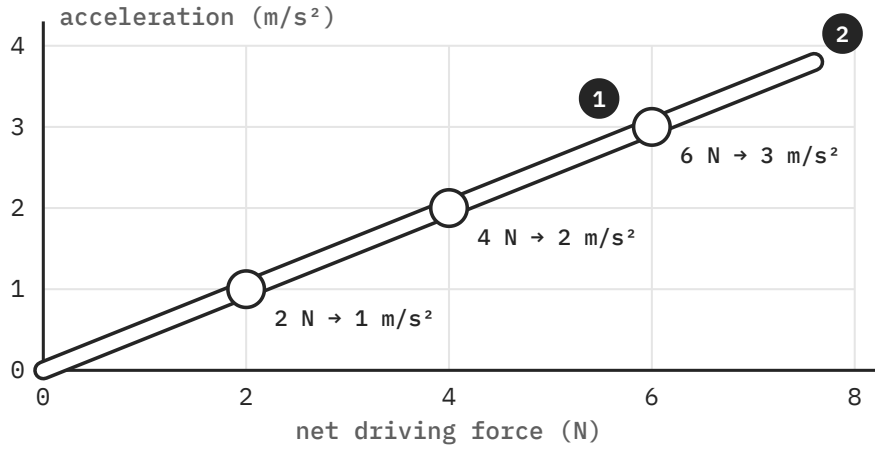
$$a \propto F \text{ and } a \propto 1/m$$
$$\rightarrow a = F \div m$$

slope of a vs. F = 1/m

| | |
|---|---|
| force | newtons (N) |
| a | m/s² |

## SECTION C COLOR IT · PLOT THE PATTERN

*Ideal 2 kg system, illustrative.*




☐ **1 · PENCIL** Data points

☐ **2 · GREEN** Trend line through zero

## SECTION D WORKED EXAMPLE

In an ideal 2 kg system, 2 N gives 1 m/s². Predict the acceleration at 4 N with the same mass.

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| F = 2 N → a = 1 m/s²<br>new F = 4 N<br>same mass | a ∝ F (same mass) | 4 N ÷ 2 N = 2× the force<br>a = 2 × 1 m/s² | **2 m/s²**<br>acceleration doubles |

## SECTION E THE 2026-XJ THREAD

DAY 12 OF 22 · NEW TOOL: NET FORCE

A bigger net force gives a bigger acceleration, in proportion. That's the rule analysts use to ask how hard XJ would have to be pushed to change course.

*Unit question: How fast is it moving, and where will it arrive?*

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Keep the mass fixed and change the force

A proportional relationship and Newton's second law

Yesterday we changed mass while trying to keep the net force fixed. Today we reverse that comparison. We keep the total moving mass the same and change the force that drives the motion. Looking at both tests helps us see how force, mass, and acceleration fit together.

In the cart-and-hanger setup, moving masses from the cart to the hanger can increase the hanging weight while keeping total moving mass unchanged. We do not add new masses to the whole system during this comparison. The cart and hanger are both included in the total. That system boundary is what makes “same mass” a meaningful claim.

For a fixed mass, acceleration is proportional to net force. Proportional means that multiplying the net force by a factor multiplies acceleration by that same factor. Doubling the net force doubles acceleration. Tripling it triples acceleration. This comparison assumes that mass stays fixed and that we have correctly included opposing forces.

The ideal **2 kg system** in **Section C** accelerates at **1 m/s² under 2 N**, at **2 m/s² under 4 N**, and at **3 m/s² under 6 N**. These points form a straight line through zero on an acceleration-versus-net-force graph. Zero net force gives zero acceleration. It does not necessarily mean zero velocity.

The example in **Section D** begins with 2 N producing 1 m/s² and asks what happens at 4 N. Compare the forces: **4 N ÷ 2 N = 2**. The force doubled, so **new acceleration = 2 × 1 m/s² = 2 m/s²**. We can check with $\boldsymbol{a} = \boldsymbol{F}_{\mathbf{net}} \div \boldsymbol{m} = \mathbf{4\ N} \div \mathbf{2\ kg} = \mathbf{2\ m/s^2}$.

This combines the two lab patterns into Newton's second law: $\boldsymbol{F}_{\mathbf{net}} = \boldsymbol{ma}$. Acceleration increases with net force and decreases with mass when the other quantity is fixed. On the graph of acceleration against net force, slope is **slope = $\mathbf{1/m}$**. For 2 kg, that slope is **0.5 m/s² per newton**: every additional newton adds half a meter per second of velocity change each second.

Real data may sit below the ideal line when plotted against the driving pull because friction reduces the net force. A bigger hanging weight is not automatically the same as a bigger net force by the same ratio. I want us to distinguish the pull we supply from the combined effect of all external forces. Use measured graph slopes for acceleration and describe the limits of the experiment.

For our fictional asteroid, 2026-XJ, this relationship lets us connect a proposed net push with a predicted acceleration. Knowing the mass and net force gives us a reasoned estimate of how its velocity would change. The prediction is only as sound as the force information and assumptions going into it.

# F = ma in practice

**LEARNING TARGET · R7** I can solve with F = ma — always finding the NET force first.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Net force** | Combine opposite forces: subtract and keep the direction of the larger one. |
| **F = ma** | Uses the *net* force, never just one pull. |
| **Direction** | Acceleration points the same way as the net force. |
| **Balanced** | Net force 0 N → a = 0 m/s² (still or constant velocity). |

## SECTION B KEY EQUATION + UNITS

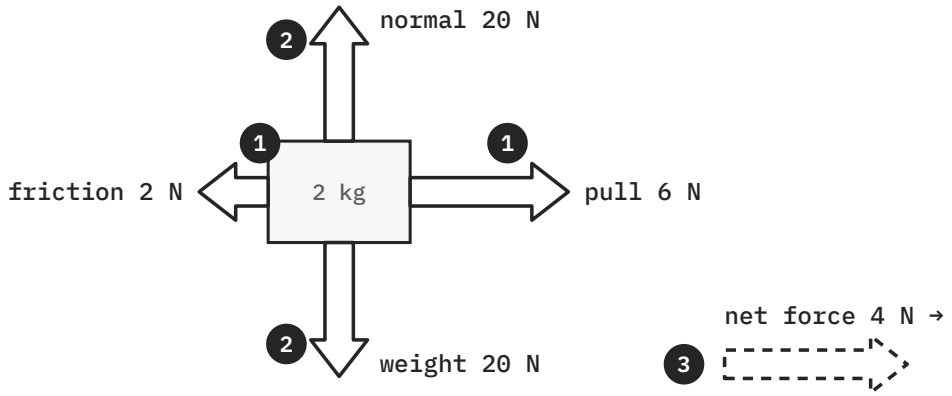
$$\Sigma F_x = m\ a_x$$

$$\Sigma F_x = F_{right} - F_{left}$$

| | |
|---|---|
| F | newtons (N) |
| m · a | kg · m/s² (1 N = 1 kg·m/s²) |

## SECTION C COLOR IT · THE ARROWS COMPETE

*Vertical forces balance.*




- ☐ **1 · ORANGE** — Horizontal forces
- ☐ **2 · YELLOW** — Vertical forces (balanced)
- ☐ **3 · ORANGE** — Net force (thick arrow)

## SECTION D WORKED EXAMPLE

A 4 kg cart has 12 N pulling right and 4 N of friction left. Find the acceleration.

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| m = 4 kg<br>12 N right<br>4 N left | $a = F_{net} \div m$ | $F_{net} = 12 - 4 = 8$ N right<br>$a = 8 \div 4$ | **2 m/s² right**<br>with the net force |

## SECTION E THE 2026-XJ THREAD

DAY 13 OF 22 · NEW TOOL: F = MA

F = ma links it all: from XJ's mass and the net force on it, we can calculate its acceleration, and from that, how its velocity will change.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Find the net force before dividing by mass

Using Newton's second law with opposing forces

A strong pull does not act alone when friction is also present. To predict acceleration, we need the combined effect of all the forces on the object. Newton's second law uses net force. I want that force sum to appear before the division by mass, so the calculation shows which interactions were included.

Choose an axis and a positive direction. With right positive, a rightward force enters the sum as positive and a leftward force as negative. Along the horizontal axis, $\Sigma F_x = ma_x$. The symbol Σ means sum. For two opposing forces, this becomes $\Sigma F_x = F_{right} - F_{left}$.

Compare the horizontal arrows in **Section C**. They act on a **2 kg cart** pulled right by **6 N**, with **2 N of friction left**. Its horizontal net force is **6 N – 2 N = 4 N** right. The 20 N upward normal force balances the 20 N downward weight. Those vertical forces do not contribute to the horizontal sum, and their own vertical sum is zero.

The thick net-force arrow summarizes the two horizontal forces. It is not an additional interaction to add again. If we counted the pull, friction, and net-force arrow as three separate forces, we would count the same imbalance twice. The diagram and force equation are two ways to describe the same situation.

The example in **Section D** uses a **4 kg cart**, a **12 N pull right**, and **4 N of friction left**. First combine the forces: $F_{net} = 12\ N - 4\ N = +8\ N$. Then use $a = F_{net} \div m$, giving $a = 8\ N \div 4\ kg = 2\ m/s^2$ right. Dividing 12 N by 4 kg would ignore friction.

Acceleration points in the direction of the net force. Its size tells us the velocity change each second. The unit check agrees: $1\ N = 1\ kg \cdot m/s^2$, so newtons divided by kilograms give m/s². We can also check backward: $(4\ kg)(2\ m/s^2) = 8\ N$, matching the net force we found.

A rightward acceleration does not prove the cart is moving right. If it already moves right, it speeds up. If it moves left, it initially slows down. We need its starting velocity to describe that part of the motion. If the forces balance, acceleration is zero; the cart can remain still or keep a constant velocity.

For our fictional asteroid, 2026-XJ, mass and net force determine acceleration. Then the existing velocity and elapsed time tell us how that acceleration changes the motion. A careful force sum is the first link in that chain. It keeps a missing or double-counted interaction from changing the entire forecast.

# Newton's third law

**LEARNING TARGET · R8** I can identify 3rd-law force pairs and explain why they never cancel.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Interaction pair** | When A pushes B, B pushes A at the same time. |
| **Equal and opposite** | Same size, same type of force, opposite directions. |
| **Different receivers** | One force acts on A, its partner acts on B — so they never cancel. |
| **Effects differ** | Equal forces, different masses → different accelerations. |

## SECTION B KEY RULE

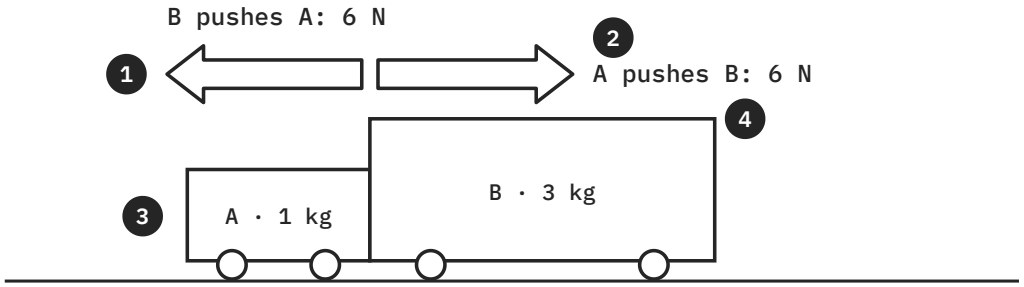
$$F_{A \text{ on } B} = -F_{B \text{ on } A}$$

swap the two object names

`force` newtons (N)
`–` opposite direction

## SECTION C COLOR IT · ONE INTERACTION, TWO FORCES

*Each arrow belongs to a different cart.*




☐ **1 · ORANGE** Force on A
☐ **2 · ORANGE** Force on B
☐ **3 · PENCIL** Cart A: label it
☐ **4 · PENCIL** Cart B: label it

## SECTION D WORKED EXAMPLE

Earth pulls an apple downward. Name the third-law partner force and the object that receives it.

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| Earth → apple<br>gravity, down | swap the object names | apple → Earth<br>gravity, up | **Apple pulls Earth up**<br>Earth receives it |

## SECTION E THE 2026-XJ THREAD

DAY 14 OF 22 · NEW TOOL: INTERACTION PAIRS

When Earth pulls on XJ, XJ pulls on Earth just as hard. The forces are equal, but XJ's far smaller mass means its motion is the one that changes noticeably.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# A force pair belongs to two different objects

Newton's third law and the receiver of each force

When you push on a wall, the wall pushes on you at the same time. These two forces come from one interaction. Newton's third law says the forces have equal sizes and opposite directions. The detail I want you to notice is where they act: your push acts on the wall, and the wall's push acts on you.

We can name a pair by swapping the two objects. If A pushes B, B pushes A. The relationship is $\boldsymbol{F}_{\mathbf{A\ on\ B}} = -\boldsymbol{F}_{\mathbf{B\ on\ A}}$. The minus sign means opposite directions. The force type is the same for both members of a pair. This is not a delay in which one force happens first and the partner follows later.

Find which cart receives each arrow in **Section C**. There, cart A has mass **1 kg** and cart B has mass **3 kg**. Their interaction pushes A left with **6 N** and B right with **6 N**. The equal-length orange arrows show equal force sizes. One arrow belongs to A and the other belongs to B. Putting both on A's diagram would describe the wrong receivers.

Equal force sizes do not mean equal accelerations. If these are the only horizontal forces, A has an acceleration of **6 N ÷ 1 kg = 6 m/s²** left. B has **6 N ÷ 3 kg = 2 m/s²** right. The smaller mass changes velocity more quickly under the same force size. The different effects do not change the equality of the interaction forces.

The apple example in **Section D** asks for the partner of Earth's downward gravitational pull on an apple. Swap the names: the apple pulls gravitationally on Earth. Earth receives that partner force, directed toward the apple. Near the apple, that is upward. Earth's much larger mass means its acceleration from this interaction is extremely small compared with the apple's.

The pair does not cancel in the force sum for the apple, because only one member acts on the apple. That is the meaning of the notes' reminder that pair forces never cancel on one object. If we choose a combined system containing both objects, the pair is internal and cancels in the combined force sum. The individual objects can still accelerate relative to each other.

Also, two balanced forces on one object are not automatically a third-law pair. Weight downward and table support upward both act on a resting book. Their partners are the book's pull on Earth and the book's push on the table. Naming the source and receiver separates an interaction pair from forces that happen to balance.

For our fictional asteroid, 2026-XJ, Earth and the asteroid pull on each other equally. Their masses differ, so their accelerations differ. I want each force placed on the correct object's diagram before we use it to explain a change in motion.

# Free-body diagrams, part 1

**LEARNING TARGET · S5** I can draw a complete FBD — every force, with arrows sized to strength.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Free-body diagram** | Only the forces acting *on* one chosen object. |
| **The dot / box** | Replace the object with a simple shape; draw arrows from it. |
| **Source** | Every arrow needs one: rope, floor, table, Earth. |
| **No "motion force"** | Motion is not an interaction. Leave velocity off the diagram. |

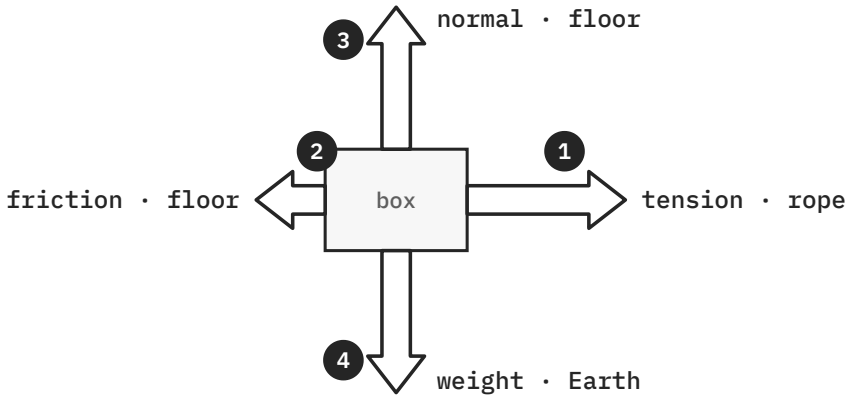
## SECTION B SIX FBD RULES

1 just the object (a dot)
2 one arrow per force
3 arrow points the force's way
4 length ∝ strength
5 label each: $F_g$, $F_N$, $F_f$…
6 only forces ON the object

**routine** choose · list · draw

## SECTION C COLOR IT · A BOX PULLED RIGHT

*Rough level floor, rope pulls right.*




- ☐ **1 · ORANGE** Tension (T)
- ☐ **2 · ORANGE** Friction (f)
- ☐ **3 · ORANGE** Normal force (N)
- ☐ **4 · ORANGE** Weight (W)

## SECTION D WORKED EXAMPLE

A 20 N lamp hangs at rest from one vertical rope. Draw and label its forces.

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| weight = 20 N<br>at rest<br>one rope | sources: Earth, rope | weight 20 N down<br>tension up, balances it | **Tension 20 N up**<br>equal-length arrows |

## SECTION E THE 2026-XJ THREAD

DAY 15 OF 22 · NEW TOOL: FREE-BODY DIAGRAMS

A free-body diagram puts XJ on a dot and shows every force acting on it. For an object in space the list is short, which is what makes it predictable.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Draw only the forces on your chosen object

A clear free-body diagram and balanced lamp forces

A free-body diagram is a simple picture of the forces acting on one chosen object. It does not need to show every detail of the scene. In fact, leaving out extra detail helps us see the interactions. Begin by deciding which object is receiving all the forces you will draw.

Use the six drawing rules in **Section B**. Replace that object with a dot or simple box. List its interactions, then draw one arrow for each force. Each arrow points in the force's direction. Its length shows the force's relative strength, using one consistent scale. Label the arrow with the force name or symbol and be able to name its source. That is the choose, list, draw routine on your notes page.

The box in **Section C** is pulled right by a rope across a rough level floor. The rope supplies rightward tension. The floor supplies leftward friction and upward normal force. Earth supplies downward weight. All four forces act on the box. The floor can supply two different contact forces because normal force and friction describe different parts of the interaction.

There is no separate “motion force.” Motion describes what the box is doing; it is not another object pushing or pulling it. Today's force diagram therefore leaves velocity off. A velocity arrow can be shown in a separate motion sketch when useful, but it must not be mistaken for an extra force. The orange arrows are reserved for the interactions.

In **Section D**, a lamp with **20 N of weight** hangs at rest from one vertical rope. Choose the lamp as the object. Earth pulls down with 20 N, and the rope pulls up with tension. There are two forces on the lamp in this model. Its being at rest does not mean no forces act.

Because the lamp remains at rest, vertical acceleration is zero and the forces balance. Choose up as positive. Then $\boldsymbol{T} - \mathbf{20\ N} = \mathbf{0}$, so $\boldsymbol{T} = \mathbf{20\ N}$. Draw equal-length arrows: 20 N down for weight and 20 N up for tension. Their sum is zero, matching the unchanged motion.

Those balanced forces are not a third-law pair. Both act on the lamp, while the partners act on different receivers: the lamp pulls on Earth and on the rope. Those partner forces belong on diagrams for Earth and the rope. Also, the net force is the sum of the two arrows, not a third arrow representing a new interaction.

For our fictional asteroid, 2026-XJ, we use the same method: choose the asteroid, identify the bodies interacting with it, and draw the forces on it. A clear diagram makes it easier to spot a missing force or an arrow that belongs to another object. That is how a simple sketch supports a reliable calculation.

# Free-body diagrams, part 2

**LEARNING TARGET · R9** I can find net force, then acceleration, straight from the FBD — inclines included.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Normal force** | Points perpendicular to (straight away from) the surface — on a ramp, it is tilted. |
| **Weight** | Always points straight down, toward Earth. |
| **Component** | The downhill part of weight. A way to calculate — not an extra force. |
| **Accel. direction** | Same direction as the net force. |

## SECTION B KEY EQUATIONS + UNITS

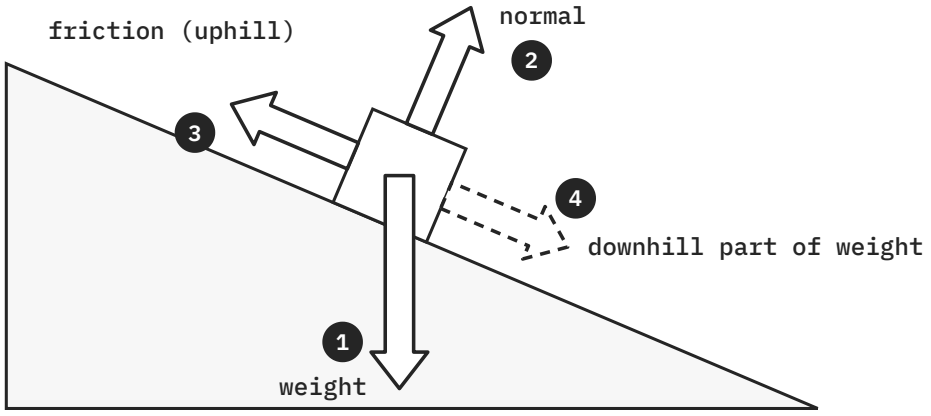
$$\Sigma F_x = m\,a_x \cdot \Sigma F_y = m\,a_y$$

down ramp: $mg \sin\theta$
into ramp: $mg \cos\theta$

+x down the ramp
+y out of the surface

## SECTION C COLOR IT · FORCES ON A RAMP

*Don't double count weight.*




☐ **1 · ORANGE** Weight (W)
☐ **2 · ORANGE** Normal force (N)
☐ **3 · ORANGE** Friction (f)
☐ **4 · ORANGE** Downhill part (dashed)

## SECTION D WORKED EXAMPLE

A 2 kg block has a 7 N downhill weight component and 3 N of friction uphill. Find its acceleration.

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| m = 2 kg<br>7 N downhill<br>3 N uphill | $a = F_{net} \div m$ | $F_{net} = 7 - 3 = 4$ N downhill<br>$a = 4 \div 2$ | **2 m/s² downhill** |

## SECTION E THE 2026-XJ THREAD

DAY 16 OF 22 · NEW TOOL: FBD → ACCELERATION

From the diagram, the net force gives XJ's acceleration and its direction. Get the net force's direction right and you know which way its path will bend.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Use the force diagram to find acceleration

Ramp directions, weight components, and net force

On a level table, normal force points up and weight points down. On a ramp, the normal force tilts with the surface, but weight still points straight down toward Earth. I want us to keep the physical force directions clear before choosing axes for the calculation.

A useful choice is a positive axis downhill along the ramp and a second positive axis straight out from the surface. These axes match the block's allowed motion and the direction of the normal force. Choosing tilted axes does not tilt gravity. It changes how we describe the parts of that same downward force.

Weight has a component along the ramp and a component into the ramp. If the ramp is at angle $\theta$ above horizontal, the downhill component has size $\boldsymbol{mg}\ \mathbf{sin}\ \boldsymbol{\theta}$, and the inward component has size $\boldsymbol{mg}\ \mathbf{cos}\ \boldsymbol{\theta}$. These are parts of weight, not additional interactions. We can use the parts instead of the full weight when adding forces along the tilted axes.

Follow the arrows in **Section C**. They show weight downward, normal force outward from the ramp, and friction uphill. It also shows the downhill part of weight with a dashed arrow. That dashed arrow helps with the calculation. Do not count both the full weight and its components as separate forces, or weight would be included twice.

In **Section D**, the downhill weight component is already given as **7 N**. Friction acts uphill with **3 N**, and the block has mass **2 kg**. Because the component is supplied, no trigonometry is needed to solve this example. With downhill positive, $\boldsymbol{F_{net}} = \mathbf{7\ N} - \mathbf{3\ N} = +\mathbf{4\ N}$ along the ramp.

Now apply Newton's second law: $\boldsymbol{a} = \boldsymbol{F_{net}} \div \boldsymbol{m} = \mathbf{4\ N} \div \mathbf{2\ kg} = \mathbf{2\ m/s^2}$ downhill. The direction matches the net force. If the block is already moving downhill, it speeds up. If it is moving uphill, the same downhill acceleration would initially slow that motion. Acceleration direction and current velocity direction are different pieces of information.

We also check the direction perpendicular to the ramp. For a block staying in contact with a straight ramp, there is no acceleration through the surface. If weight and normal force are the only forces with perpendicular components, the normal force balances the inward part of weight. It does not have to equal the whole weight. In general, we use $\boldsymbol{\Sigma F_x} = \boldsymbol{ma_x}$ and $\boldsymbol{\Sigma F_y} = \boldsymbol{ma_y}$ for the two chosen axes.

For our fictional asteroid, 2026-XJ, a force diagram likewise gives the net force and acceleration direction. The existing velocity still matters when describing the resulting path. Useful axes simplify the calculation, while a clear distinction between actual forces and components keeps the model consistent.

# Friction: before and during sliding

**LEARNING TARGET** I can distinguish adjustable static friction from kinetic friction.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Static friction** | Before sliding. Adjusts to match the pull, up to a limit. |
| **Maximum static** | The greatest static friction possible. Pull past it and sliding starts. |
| **Kinetic friction** | During sliding. Use the supplied value. |
| **Coefficient (μ)** | When given: kinetic friction = μ × normal force. |

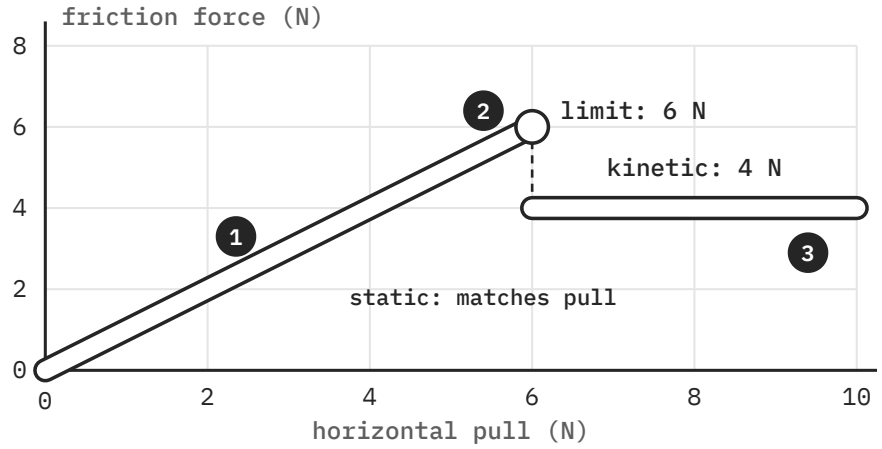
## SECTION B KEY EQUATIONS + UNITS

$$\text{at rest: } f_s = \text{pull } (\le \text{limit})$$
$$\text{sliding: } f_k = \mu \times F_N$$

| | |
|---|---|
| friction | newtons (N) |
| μ | no unit |

## SECTION C COLOR IT · BEFORE AND DURING SLIDING

*Example crate: limit 6 N, kinetic 4 N.*




- ☐ **1 · ORANGE** — Static friction: matches pull
- ☐ **2 · ORANGE** — The limit (dashed)
- ☐ **3 · ORANGE** — Kinetic friction

## SECTION D WORKED EXAMPLE

A 2 kg crate slides right. Pull = 10 N right; kinetic friction = 4 N left. Find its acceleration.

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| m = 2 kg<br>pull 10 N right<br>$f_k$ 4 N left | $a = F_{net} \div m$ | $F_{net} = 10 - 4 = 6$ N right<br>$a = 6 \div 2$ | **3 m/s² right** |

## SECTION E THE 2026-XJ THREAD

DAY 17 OF 22 · NEW TOOL: FRICTION

Friction needs surfaces rubbing together. In space there are none to slow XJ down, one more reason it keeps its speed between observations.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Friction adjusts before sliding begins

Static friction, its limit, and kinetic friction

A gentle push on a crate may leave it at rest. A stronger push may still leave it at rest. That does not mean the pushing force disappeared. Static friction can adjust to balance the pull, up to a limit. Today I want us to decide whether the surfaces are sliding before choosing a friction value.

Static friction acts when the contact surfaces are not slipping relative to each other. In the simple horizontal example, it matches the pull while the crate remains at rest. If the pull is 2 N, static friction can be 2 N backward. If the pull rises to 5 N, it can rise to 5 N, provided the maximum available friction has not been exceeded.

Find where the rising line stops in **Section C**. The graph uses a **6 N static-friction limit**. The rising line shows friction matching the pull up to that limit. A pull greater than the limit cannot be balanced by static friction in this example, so sliding begins. The 6 N value is the greatest available static friction, not the friction that acts during every smaller pull.

During sliding, we use kinetic friction. The illustrative crate has **4 N of kinetic friction**, shown by the lower horizontal part of the graph. The friction value changes because the contact is in a different state. We should not continue using the maximum static value after the crate is sliding.

The example in **Section D** gives a **2 kg crate sliding right**, a **10 N pull right**, and **4 N of kinetic friction left**. First find the net force: $\boldsymbol{F_{\text{net}}}$ **= 10 N − 4 N = 6 N** right. Then $\boldsymbol{a}$ **=** $\boldsymbol{F_{\text{net}}}$ **÷** $\boldsymbol{m}$ **= 6 N ÷ 2 kg = 3 m/s²** right. The crate is moving right and accelerating right, so it speeds up.

Sometimes kinetic friction is not given directly. If a coefficient $\mu$ and normal force are supplied, use $\boldsymbol{f_{\text{k}} = \mu F_{\text{N}}}$. The coefficient has no unit, so the answer has the same force unit as the normal force: newtons. For instance, a coefficient of 0.20 with a 20 N normal force would give **(0.20)(20 N) = 4 N**. Find the normal force from the situation rather than assuming a value.

Friction opposes slipping, or the tendency to slip, at the contact. Its direction must follow that contact behavior. It is not a rule that friction always points opposite an object's overall motion in every possible situation. A shoe can receive forward static friction from the ground while a person walks forward.

For our fictional asteroid, 2026-XJ, there is no rubbing surface in empty space to apply this kind of friction. Gravity can still change its speed or direction, and atmospheric drag would be a different interaction inside an atmosphere. I want us to include the forces whose sources are actually present.

# Equilibrium: find the missing balance

**LEARNING TARGET** I can find a missing force when acceleration is zero.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Equilibrium** | Acceleration is zero. The forces sum to zero in every direction. |
| **Two kinds** | At rest *or* moving at constant velocity. |
| **Missing force** | Whatever is needed to make each direction balance. |
| **Careful** | Zero velocity for an instant does not guarantee equilibrium. |

## SECTION B KEY CONDITION

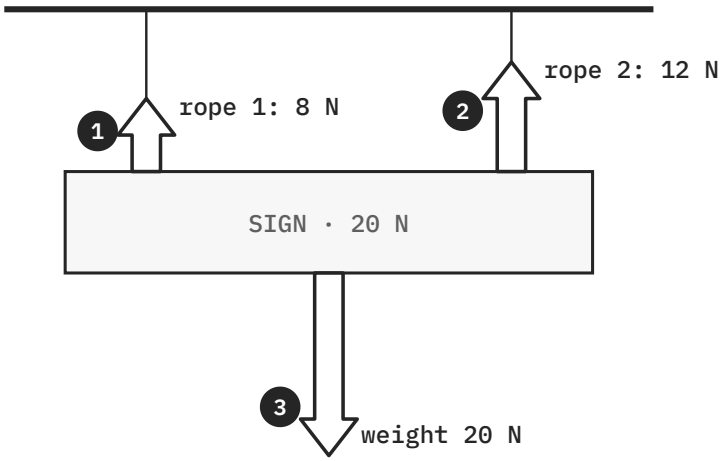
$$\Sigma F = 0$$

up = down
right = left

Σ "sum of"
force newtons (N)

## SECTION C COLOR IT · SUPPORTS MUST TOTAL THE WEIGHT

*8 N + 12 N = 20 N.*




☐ **1 · ORANGE** Rope 1 tension
☐ **2 · ORANGE** Rope 2 tension
☐ **3 · ORANGE** Weight (W)

## SECTION D WORKED EXAMPLE

A 40 N sign hangs at rest from two vertical ropes. One tension is 15 N. Find the other.

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| weight 40 N<br>rope 1 = 15 N<br>at rest | up = down | $15 + T_2 = 40$<br>$T_2 = 40 - 15$ | **25 N up** |

## SECTION E THE 2026-XJ THREAD

DAY 18 OF 22 · NEW TOOL: EQUILIBRIUM

If the forces on XJ balanced, its acceleration would be zero and its velocity would stay constant. Analysts check for balance before assuming steady motion.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Find what is needed to make the forces balance

Equilibrium and missing support forces

A sign hanging at rest has forces acting on it. The reason it stays at rest is that those forces balance. Equilibrium means zero acceleration, so the force sum is zero in every direction. Today we will use that condition to find a missing force rather than assume every support carries an equal share.

The shorthand is $\Sigma F = 0$. Along a vertical axis, that means the upward forces total the downward forces. Along a horizontal axis, rightward and leftward totals must match. Balance in one direction does not automatically prove balance in another. We check each direction that has forces acting along it.

Compare the upward arrows in **Section C**. The diagram shows a **20 N sign** supported by two vertical ropes. One pulls up with **8 N**, and the other pulls up with **12 N**. Their combined support is **8 N + 12 N = 20 N**, matching the downward weight. The two ropes do not need equal tensions. Their sum needs to provide the required upward force.

The example in **Section D** uses a **40 N sign**. It remains at rest, and one vertical rope has a tension of **15 N**. Call the other tension $T_2$. Because up equals down, $15\ \text{N} + T_2 = 40\ \text{N}$. Subtract the known support from the weight: $T_2 = 40\ \text{N} - 15\ \text{N} = 25\ \text{N}$ upward.

Check the result by adding it back: $15\ \text{N} + 25\ \text{N} - 40\ \text{N} = 0\ \text{N}$. That agrees with zero acceleration. Dividing the weight equally between the ropes would have given 20 N each, but that would ignore the supplied 15 N tension. I want the given evidence, not a picture that merely looks symmetrical, to determine the calculation.

These ropes are vertical, so their entire tension acts upward. If the ropes were angled, only their vertical components would support the weight and we would need another step. Naming the directions keeps a simple balance rule from being used in a situation where it no longer applies directly.

Equilibrium can also describe an object moving with constant velocity. Zero net force means velocity does not change; it does not require the velocity to be zero. On the other hand, zero velocity at one instant does not prove equilibrium. A tossed object can be momentarily stopped at its highest point while gravity still gives it downward acceleration.

For our fictional asteroid, 2026-XJ, balanced forces would mean constant velocity. They would not automatically bring the asteroid to a stop. Before assuming steady motion, we need a reason to believe the net force is zero or small enough for our chosen approximation. The force balance and the motion model should support the same explanation.

# Workshop: find and fix the reasoning

**LEARNING TARGET** I can use a picture to identify a mistake and explain a correction.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Distance vs. displacement** | Trace the whole path vs. join start to finish. |
| **Net force first** | Combine all forces before dividing by mass. |
| **Sign = direction** | A minus sign is not "slow." Compare sizes for speed. |
| **Pair forces** | Act on different objects, so they never cancel. |

## SECTION B KEY ROUTINE

1. Identify the first mistake
2. Repair it with a drawing + rule
3. Recheck on a new problem

**evidence** point to a graph or diagram
**reason** name the physics rule

## SECTION C COLOR IT · FOUR COMMON MISTAKES

*Lightly shade each strip.*

**1** **CLAIM** ✗ "3 m right then 3 m back — the distance is zero."

**2** **FIX** ✓ Distance = 3 + 3 = 6 m. *Displacement* is 0 m: the endpoints match.

**1** **CLAIM** ✗ "8 N pull on a 2 kg cart: 8 ÷ 2 = 4 m/s²."

**2** **FIX** ✓ Net force first: 8 − 2 = 6 N. Then 6 ÷ 2 = 3 m/s².

**1** **CLAIM** ✗ "−2 → −6 m/s is slowing, because it's negative."

**2** **FIX** ✓ Speed grows from 2 to 6 m/s: speeding up. The minus sign is direction.

**1** **CLAIM** ✗ "The force pair cancels, so nothing moves."

**2** **FIX** ✓ Each pair force acts on a different cart. Draw one on each diagram.

☐ **1 · PENCIL** Mistaken claims: cross out

☐ **2 · CODE COLOR** The fixes: color by quantity

## SECTION D WORKED EXAMPLE

A 3 kg cart has 9 N right and 3 N left. A student writes 9 ÷ 3 = 3 m/s². Correct it.

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| m = 3 kg<br>9 N right<br>3 N left | $a = F_{net} \div m$ | $F_{net} = 9 - 3 = 6$ N right<br>$a = 6 \div 3$ | **2 m/s² right**<br>missed: net force first |

## SECTION E THE 2026-XJ THREAD

DAY 19 OF 22 · NEW TOOL: CHECKING REASONING

A forecast is only as good as its reasoning. Before the desk signs off on XJ, every step is checked: signs, starting position, units, and the right model.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Repair the reasoning, not just the answer

Finding the first mistake and checking the correction

When an answer is wrong, changing the final number does not explain what happened. I want us to find the first step where the reasoning stopped matching the situation. Then we can repair that step and follow the correction through the rest of the work. The pictures on today's notes page make four common mistakes visible.

Check the first strip in **Section C**. The claim confuses distance and displacement. A trip goes **3 m right and 3 m back**. The total distance is **3 m + 3 m = 6 m**. Displacement is **0 m** because the starting and finishing positions match. A route sketch shows both facts: tracing the full path gives distance, while joining the endpoints gives displacement.

The second claim uses an applied force as if it were the net force. An **8 N pull** acts on a **2 kg cart**, with **2 N opposing it**. Before dividing by mass, find $F_{net} = 8\ N - 2\ N = 6\ N$. Then $a = 6\ N \div 2\ kg = 3\ m/s^2$. The missing subtraction was the first wrong step.

The third claim calls a change from **−2 m/s to −6 m/s** slowing down. The minus sign indicates direction. Speed is the size of velocity, so it grows from 2 m/s to 6 m/s. The object is speeding up in the negative direction. A velocity-time graph would move farther below zero; that does not mean its speed is getting smaller.

The fourth claim says a third-law pair cancels so nothing moves. Each member acts on a different object. Draw one force on each object's separate force diagram. They cannot cancel in the force sum for one receiver. If we combine both objects into one system, the pair is internal, but the objects can still change their motion relative to each other.

The example in **Section D** gives another force-sum repair. A **3 kg cart** has **9 N right** and **3 N left**. Dividing the 9 N pull by mass ignores the other force. The correction is $F_{net} = 9\ N - 3\ N = 6\ N$ right, followed by $a = 6\ N \div 3\ kg = 2\ m/s^2$ right. Check backward: $(3\ kg)(2\ m/s^2) = 6\ N$, which matches the net force.

A useful explanation points to evidence and names a rule. The drawing can show a missing force, a return trip, or the receiver of an interaction. The equation then shows the correction. Keep units with the numbers and check whether the sign describes the chosen positive direction. Repeating the same mistaken substitution in another formula is not an independent check.

For our fictional asteroid, 2026-XJ, one missed starting position or incorrect sign can affect every later prediction. Before trusting the forecast, trace it back through its evidence and model. Showing the first repaired step makes a revision understandable instead of leaving someone to guess why the final answer changed.

# Build a complete prediction

**LEARNING TARGET** I can connect forces, velocity change, and final position.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **The chain** | Net force → acceleration → velocity → position. |
| **Constant net force** | Gives constant acceleration: a straight v–t line. |
| **Zero net force** | Velocity stays the same; position still changes. |
| **Range** | A central value with limits. Not a guarantee. |

## SECTION B KEY EQUATIONS

$$a = F_{net} \div m$$
$$v_f = v_i + a \times t$$
$$\Delta x = \tfrac{1}{2}(v_i + v_f) \times t$$
$$x_f = x_i + \Delta x$$

`N, kg` force, mass

`m/s², m/s, m` a, v, x

## SECTION C COLOR IT · ONE CHAIN, FOUR LINKS

*2 kg cart, from rest at 1 m, for 2 s.*

**1 NET FORCE**
`6 − 2 = 4 N`
pull minus friction

→

**2 ACCEL.**
`4 ÷ 2 = 2 m/s²`
$F_{net} \div m$

→

**3 VELOCITY**
`0 + 2×2 = 4 m/s`
gain 2 m/s each second

→

**4 POSITION**
`1 + 4 = 5 m`
Δx = 2 m/s avg × 2 s

Report the range: **`5 ± 1 m → 4 to 6 m`** – the central value is not guaranteed.

- ☐ **1 · ORANGE** Force
- ☐ **2 · GREEN** Acceleration (speeding up)
- ☐ **3 · PURPLE** Velocity
- ☐ **4 · BLUE** Position

## SECTION D WORKED EXAMPLE

A cart has zero net force and already moves at +2 m/s. After 3 s, find its velocity and displacement.

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| $F_{net}$ = 0 N<br>v = +2 m/s<br>t = 3 s | a = 0 → v constant<br>Δx = v × t | v stays +2 m/s<br>Δx = 2 × 3 | **+2 m/s, +6 m** |

## SECTION E THE 2026-XJ THREAD

DAY 20 OF 22 · NEW TOOL: THE FULL CHAIN

The chain is complete: net force → acceleration → velocity → position. That is the whole route from the data on XJ to a forecast of where it will arrive.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Connect the forces to the final position

The complete chain from net force to a motion prediction

We have learned how forces change velocity and how velocity changes position. Today we connect those ideas in one calculation. I want you to keep the same object, positive direction, and time interval throughout the chain. Each result supplies information needed for the next step.

Follow the four links in **Section C**. The chain begins with a **2 kg cart** at rest at **1 m**. A **6 N pull** acts right while **2 N of friction** acts left. These forces stay constant for **2 s**. Choose right as positive. First find $F_{net}$ = **6 N − 2 N = +4 N**. Then $a = F_{net} \div m$ = **4 N ÷ 2 kg = 2 m/s²**.

The cart gains 2 m/s of velocity each second. Use $v_f = v_i + at$ to find $v_f$ = **0 m/s + (2 m/s²)(2 s) = +4 m/s**. Constant net force on a fixed mass gives constant acceleration, so its velocity-time graph would be a straight rising line.

Because the acceleration is constant, average velocity is **(0 + 4) m/s ÷ 2 = 2 m/s**. Multiply by the two seconds to find $\Delta x$ = **(2 m/s)(2 s) = +4 m**. Now include the starting position: $x_f$ = **1 m + 4 m = 5 m**. The cart's displacement is four meters, while its final position is the five-meter mark.

The diagram reports an illustrative bound of **5 ± 1 m**, meaning a range from **4 m to 6 m**. That one-meter bound is supplied for the example; it does not come from the force calculation alone. The center is not guaranteed to be the exact measured result, and the range is not a probability statement.

**Section D** tests a different case: a cart already moving at **+2 m/s** has **zero net force** for **3 s**. Zero net force gives zero acceleration, so the velocity stays at +2 m/s. Its displacement is $\Delta x$ = **(2 m/s)(3 s) = +6 m**. The cart keeps moving even though no unbalanced force acts on it.

Notice what information each step needs. Force and mass give acceleration. Starting velocity and time are needed to find final velocity. Starting position is needed to find final position. Orange identifies the force step, purple identifies velocity, and blue identifies position. Green in this particular diagram marks speeding up; acceleration itself does not always mean increasing speed.

For our fictional asteroid, 2026-XJ, this chain connects the observations and interactions to a forecast. A force calculation alone is not the whole prediction. We also need the existing motion and starting location, along with a model that fits the interval being studied.

# Transfer task: make a plan

**LEARNING TARGET** I can organize the assigned evidence and begin a defensible prediction.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Reference point** | The zero everything is measured from. Mark it on your sketch. |
| **Given** | Every value copied with its unit. |
| **Model assumption** | State it: steady velocity or constant acceleration. |
| **Range** | Read the lower and upper values before making a claim. |

## SECTION B KEY HABIT + TOOLS

quantity → numbers + units → result

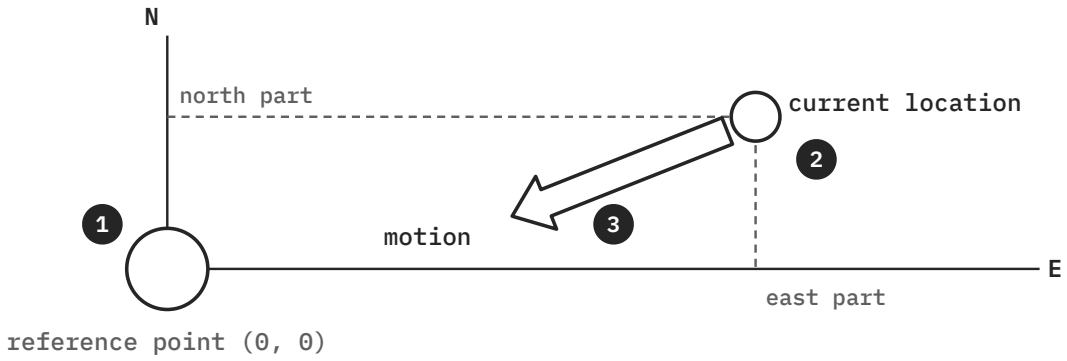
$$x_f = x_i + v \times t$$

$$v_f = v_i + a \times t$$

`one line` one quantity
`E–W, N–S` solve separately

## SECTION C COLOR IT · MAKE THE SITUATION VISIBLE

*Your sketch needs all three.*




- ☐ **1 · PENCIL** Reference point
- ☐ **2 · BLUE** Current location
- ☐ **3 · PURPLE** Motion direction

## SECTION D WORKED EXAMPLE

Check a prediction: an object starts at 2 m and moves at a steady +3 m/s for 4 s.

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| $x_i = 2$ m<br>$v = +3$ m/s<br>$t = 4$ s · steady v | $x_f = x_i + v \times t$ | change = $3 \times 4 = +12$ m<br>$x_f = 2 + 12$ | **14 m**<br>check: direction + size + start |

## SECTION E THE 2026-XJ THREAD

DAY 21 OF 22 · NEW TOOL: YOUR PLAN

Now it's your forecast. Pick the evidence on XJ that matters, choose a model, and plan the calculation that answers the unit question.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Make the situation clear before calculating

A reference frame, useful evidence, and a prediction plan

A prediction begins with a situation, not a formula. Before calculating, I want us to identify where the object starts, how it moves, and what we need to find. A clear sketch and a short list of given values can prevent mistakes that are difficult to notice after several lines of arithmetic.

Choose a reference point and mark it as zero. Name the positive direction, or both axes if the motion is two-dimensional. Find the three features of the sketch in **Section C**: a reference point, a current location, and a motion arrow. Those features do different jobs. Blue identifies location, purple identifies motion, and the reference axes remain in pencil.

Copy each given quantity with its unit. A position in meters is different from a velocity in meters per second. Time must also use units that match the velocity or acceleration. Keeping each quantity on its own line makes it easier to see whether an equation uses all the information it needs.

Next choose a model and state its assumption. If velocity stays steady, use $x_f = x_i + vt$. If constant acceleration changes velocity, $v_f = v_i + at$ describes that change. A position prediction under acceleration also needs an appropriate displacement step. Do not choose a steady-velocity model simply because it gives the shortest calculation.

The prediction in **Section D** gives a starting position of **2 m**, a steady velocity of **+3 m/s**, and an elapsed time of **4 s**. The question asks for final position. First find the change: $\Delta x = (3\ \text{m/s})(4\ \text{s}) = +12\ \text{m}$. Then add the start: $x_f = 2\ \text{m} + 12\ \text{m} = 14\ \text{m}$. Writing both steps makes it clear why the answer is not just 12 m.

Check the result against the story. Positive velocity should increase the coordinate. Four seconds at three meters per second should add twelve meters. Starting at two meters then places the object at fourteen meters. A number-line sketch can confirm both the direction and the size of that change. The units also reduce correctly to meters.

In two dimensions, solve the horizontal and vertical coordinates separately using the corresponding velocity components and the same elapsed time. Do not use the whole diagonal speed as if it were the speed along each axis. If the evidence includes a prediction range, read its lower and upper limits before making a claim; the central value alone does not describe the whole result.

For our fictional asteroid, 2026-XJ, the plan connects relevant measurements to a stated model. A classroom model may treat Earth as a fixed reference and velocity as steady over a chosen interval. Those are assumptions to explain and check. A defensible plan shows how the evidence can answer where the asteroid will be at a particular future time without claiming information the measurements do not provide.

# Transfer task: defend, revise, submit

**LEARNING TARGET** I can explain my prediction, identify limits, and improve my reasoning.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Claim** | Your prediction, stated clearly. |
| **Evidence** | Visible work: a graph, sketch, or calculation. |
| **Limitation** | What the model leaves out or assumes. |
| **Revision** | A corrected step, shown — and how it changes later results. |

## SECTION B KEY HABIT

`a correct-looking number ≠ shown reasoning`

`physics predicts; people decide`

`central value` + its range
`revision` show the fixed step

## SECTION C COLOR IT · BUILD A DEFENSE

*Point to your own work as you speak.*

**1 CLAIM**
I predict ___.

**2 EVIDENCE**
based on ___ (point to the graph, sketch, or calculation).

**3 REASONING**
because my model assumes ___ .

**4 LIMITATION**
The result is limited by ___. A range is not a probability, and a recommendation also involves cost, feasibility, and priorities.

- ☐ **1 · BLUE** Claim: your predicted position
- ☐ **2 · CODE COLOR** Evidence: color by quantity
- ☐ **3 · PENCIL** Reasoning
- ☐ **4 · PENCIL** Limitation

## SECTION D WORKED EXAMPLE

Defend a prediction: a 2 kg cart under 4 N net force, from rest at 1 m, for 2 s; model bound ± 1 m.

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| $F_{net}$ = 4 N, m = 2 kg<br>$x_i$ = 1 m, t = 2 s<br>bound ± 1 m | force → a → v → x | a = 2 m/s², v = 4 m/s<br>Δx = 4 m, $x_f$ = 5 m | **4 to 6 m**<br>assumes constant forces |

## SECTION E THE 2026-XJ THREAD

DAY 22 OF 22 · NEW TOOL: YOUR DEFENSE

The desk wants a claim it can defend: where XJ will be, how sure you are, and what your model leaves out. That is what makes a prediction trustworthy.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Explain what the forecast means and what limits it

Claim, evidence, reasoning, and a visible revision

A correct-looking number is not the same as shown reasoning. Someone reading a forecast needs to know what was predicted, which evidence supports it, and what assumptions connect the two. Today I want the explanation to be clear enough that another person could follow the calculation and understand its limits.

Follow the parts of the defense in **Section C**. A claim states the result. Evidence is the visible work supporting it: measurements, a graph, a sketch, or a calculation. Reasoning explains why the physics model connects that evidence to the claim. A limitation identifies something uncertain, assumed, or left out. These four parts work together rather than replacing one another.

The forecast in **Section D** uses a **2 kg cart** under a constant **4 N net force**. It starts at rest at **1 m**, and we predict its position after **2 s**. First find $a = 4\ \text{N} \div 2\ \text{kg} = 2\ \text{m/s}^2$. Then $v_f = 0\ \text{m/s} + (2\ \text{m/s}^2)(2\ \text{s}) = 4\ \text{m/s}$. The constant net force and fixed mass justify constant acceleration.

Average velocity is $(0 + 4)\ \text{m/s} \div 2 = 2\ \text{m/s}$, so $\Delta x = (2\ \text{m/s})(2\ \text{s}) = 4\ \text{m}$. Add the starting coordinate: $x_f = 1\ \text{m} + 4\ \text{m} = 5\ \text{m}$. The example supplies a model bound of $\pm 1\ \text{m}$, so the reported position interval is **4 m to 6 m**. The bound is part of the supplied information, not something this arithmetic creates by itself.

A defense can now connect the pieces: we predict a central position of 5 m after two seconds, with the given range from 4 m to 6 m. The evidence is the force, mass, starting conditions, and time. The reasoning uses Newton's second law and constant-acceleration motion. The result is limited by the supplied bound and the assumption that the forces stay constant throughout the interval.

A range is not a probability. This example does not tell us that some percentage of outcomes must occur inside it. Also, a model can miss a changing force even when its arithmetic is correct. I want uncertainty in the inputs and limits of the model named honestly, rather than hidden behind extra decimal places.

A revision should show the corrected step and its effect on later results. For example, if the starting coordinate were corrected from 1 m to 2 m, with everything else unchanged, the displacement would still be 4 m but the central final position would become $2\ \text{m} + 4\ \text{m} = 6\ \text{m}$. We can explain exactly why that result changed instead of simply replacing the old answer.

For our fictional asteroid, 2026-XJ, a useful forecast includes a location, future time, reference frame, supporting work, and limits. Decisions about a response also involve cost, feasibility, and priorities. Physics supplies evidence for those decisions. A clear defense shows what the model supports and what additional information people still need.